

Supplementary Material for: Computational Tool Choice Impacts CRISPR Spacer-Protospacer Detection

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1 Supplementary Materials

1.1 Tables

1.1.1 Supplementary Table S1: Tool Configuration Details

Note on BLASTn parameters: The parameters for BLASTn were specifically chosen to enable the highest possible recall for short spacers. We used `blastn-short` (sets word-size at 7, compared to 11 with default `blastn` and 28 in `megablastn`), a very high number of max target sequences (`-max_target_seqs 1000000`, allowing up to 1M reported alignments per query compared to default value of 500), the `--ungapped` flag (used by CRISPRTarget, aligns with our hamming distance focus), and `perc_identity=84` and `qcov_hsp_perc=80`. The `ungapped` flag considerably reduced total CPU time. The `perc_identity` and `qcov_hsp_perc` arguments control output size - we cannot restrict `blastn` to a specific “n” distance, so these values allow a range of distances to be reported. For example, 85% identity over at least 80% of the spacer allows for 3 mismatches for a 20bp spacer ($20 * 0.15 = 3$), but allows more for a 30bp spacer ($30 * 0.15 = 4.5$). This range exceeds our desired thresholds to capture all valid matches.

Tool	Version	Command
blastn	BLAST+ 2.16.0	<pre>makeblastdb -in {contigs_file} -dbtype nucl -out {output_dir}/contigs_blastdb && blastn -query {spacers_file} -db {output_dir}/contigs_blastdb -max_target_seqs 1000000 -out {output_dir}/blastn_output.tsv -perc_identity 84 -qcov_hsp_perc 80 -dust no -ungapped -num_threads {threads} -task blastn-short -outfmt "6 qaccver saccver nident length mismatch qlen gapopen qstart qend sstart send eval evalue bitscore"</pre>
bowtie1	1.3.1	<pre>bowtie-build --threads {threads} {contigs_file} {output_dir}/bt1_contigs_index && bowtie --threads {threads} --seedmms 3 --no-unal -f --all -v {max_mismatches} -x {output_dir}/bt1_contigs_index {spacers_file} -S {output_dir}/bowtie1_output.sam</pre>

Tool	Version	Command
bowtie2	2.5.4	<pre> bowtie2-build --large-index --threads {threads} {contigs_file} {output_dir}/contigs_bt2indx && bowtie2 -N 1 -L 15 --all --xeq --very-sensitive -x {output_dir}/contigs_bt2indx -f {spacers_file} -S {output_dir}/bowtie2_output.sam --threads {threads} </pre>
indelfree_bruteforce	BBTools 39.62	<pre> indelfree.sh subs={max_mismatches} in={spacers_file} ref={contigs_file} out={output_dir}/indelfree_bruteforce_output.s index=f threads={threads} simd </pre>
indelfree_indexed	BBTools 39.62	<pre> indelfree.sh subs={max_mismatches} in={spacers_file} ref={contigs_file} out={output_dir}/indelfree_indexed_output.sam index=t threads={threads} k=8 </pre>
mmseqs2	18.8cc5c	<pre> mkdir -p {output_dir}/tmp_spacers {output_dir}/tmp_contigs {output_dir}/tmp_mmseqs {output_dir}/tmp_mmseqs_outputs && mmseqs createdb {spacers_file} {output_dir}/tmp_spacers/mmdb && mmseqs createdb {contigs_file} {output_dir}/tmp_contigs/mmdb && mmseqs search {output_dir}/tmp_spacers/mmdb {output_dir}/tmp_contigs/mmdb {output_dir}/tmp_mmseqs_outputs/mmseqs_output {output_dir}/tmp_mmseqs --min-seq-id 0.85 --min-aln-len 17 --threads {threads} -a --search-type 3 -v 1 && mmseqs convertalis {output_dir}/tmp_spacers/mmdb {output_dir}/tmp_contigs/mmdb {output_dir}/tmp_mmseqs_outputs/mmseqs_output {output_dir}/mmseqs_output.tsv --format-mode 0 --search-type 4 --format-output query,target,nident,alnlen,mismatch,qlen,gapopen </pre>

Tool	Version	Command
mummer4	4.0.0rc1	nucmer --maxmatch --nosimplify --batch=10000000 --threads {threads} --sam-long={output_dir}/mummer4_output.sam -c 1 {contigs_file} {spacers_file}
sassy	0.1.4	sassy search --alphabet iupac --pattern-fasta {spacers_file} -k {max_mismatches} --output {output_dir}/sassy.tsv --threads {threads} {contigs_file}
strobealign	0.16.1	strobealign --eqx -k 15 -N 10000 -M 200 -t {threads} {contigs_file} {spacers_file} -o {output_dir}/strobealign_output.sam
x_mapper	1.2.0	x_mapper --reference {contigs_file} --queries {spacers_file} --out-sam {output_dir}/x_mapper_output.sam --num-threads {threads} --max-penalty 0.2 --max-penalty-span 2 --snp-penalty 1 --new-indel-penalty 1.5 --extend-indel-penalty 0.5 -Xmx28G
minimap2	2.30-r1287	minimap2 -k 8 -w 6 -P --eqx -m 10 --sam-hit-only -t {threads} -a {contigs_file} {spacers_file} -o {output_dir}/minimap2_output.sam

1.1.2 Supplementary Table S2: IMG/VR4 Contig Dataset Statistics and Filtering Steps

Summary of IMG/VR4 v1.1 high-confidence viral contigs after taxonomic filtering and length cutoff.

Step	Operation	Output contigs	Notes
1	Load IMG/VR4 v1.1 high-confidence set	5,457,198	Initial high-confidence contigs
2	Remove eukaryotic virus taxa	5,115,894	Filtered by ICTV class/order/family lists in <code>fetch_real_data.ipynb</code>
3	Remove short contigs	5,115,894	Length filter: keep contigs >1000 bp
4	Export filtered reference	5,115,894	<code>filtered_contigs.fasta</code> and <code>filtered_contig_stats.parquet</code>

Step	Operation	Output contigs	Notes
5	Build HQ benchmark subset	421,431	Stratified taxonomic subsampling by class for fraction_1 analyses

The filtered prokaryotic viral set used for sampling contains approximately 79 Gbp total sequence length, with contig lengths ranging from 1,001 to 2,473,870 bp (median 7,664 bp). The fraction_1 subset used for full real-data benchmarking contains 421,431 contigs (approximately 1.69 Gbp).

1.1.3 Supplementary Table S3: CRISPR Spacer Dataset Composition and Statistics

Detailed statistics on the iPHoP spacer dataset before and after complexity filtering.

Stage	N spacers	Length range (bp)	Median length (bp)	GC (%)	Notes
Raw iPHoP spacer set	3,882,812	25-40	34	47.6	Input dataset used in <code>spacer_inspection.ipynb</code>
Removed by complexity/ambiguity filtering	55,833	—	—	—	Removed if <code>n_frac > 0</code> , homopolymer-like base fraction ≥ 0.95 (A/C/G/T or GC), entropy ≤ 1 , or repeated 6-mer count ≥ 4
Final filtered spacer set	3,826,979	25-40	34	46.73	Exported as <code>iphop_filtered_spacers.fna</code> ; used in real-data runs

Total filtered spacer sequence length is approximately 132 Mbp. Distribution summaries are shown in Supplementary Figures S5 and S6.

1.1.4 Supplementary Table S4: Detailed Statistics for All Datasets

A. Real datasets

Dataset / Subset	Type	N Seqs	Total Size (bp)	Length Range (bp)	Mean Length	GC %
all spacers	Spacers	3,826,979	129,494,053	25-40	33.84	47.6%
fraction 0.0005	Contigs	279	7,036,591	1,060-233,373	25,220.76	48.0%
fraction 0.001	Contigs	421	9,745,671	1,700-246,318	23,148.86	48.0%
fraction 0.005	Contigs	2,107	57,059,412	1,060-618,736	27,080.88	47.2%
fraction 0.01	Contigs	4,214	123,668,270	1,060-482,156	29,347.00	47.3%
fraction 0.05	Contigs	21,071	715,073,062	1,060-505,259	33,936.36	47.4%

Dataset / Subset	Type	N Seqs	Total Size (bp)	Length Range (bp)	Mean Length	GC %
fraction 0.1	Contigs	42,143	1,504,536,616	1,060-596,785	35,700.75	47.5%
fraction 1	Contigs	421,431	18,870,643,188	1,060-711,471	44,777.54	47.5%

B. Simulated datasets

Dataset / Subset	Type	N Seqs	Total Size (bp)	Length Range (bp)	Mean Length	GC %
ns 100000 nc 10000	Contigs	10,000	802,832,723	10,067- 149,921	80,283.27	49.0%
ns 100000 nc 10000	Spacers	100,000	3,201,222	25-40	32.01	49.0%
ns 100000 nc 20000	Contigs	20,000	1,604,143,975	10,014- 149,390	80,207.20	49.0%
ns 100000 nc 20000	Spacers	100,000	3,200,186	25-40	32.00	49.0%
ns 100 nc 5000	Contigs	5,000	380,461,323	2,621-149,596	76,092.26	46.0%
ns 100 nc 5000	Spacers	100	3,507	25-45	35.07	49.4%
ns 3826979 nc 421431 real baseline	Contigs	421,431	21,470,981,225	2,006-100,000	50,947.80	46.0%
ns 3826979 nc 421431 real baseline	Spacers	3,826,979	129,494,053	25-40	33.84	47.6%
ns 500000 nc 100000	Contigs	100,000	28,002,479,279	10,140- 549,926	280,024.79	49.0%
ns 500000 nc 100000	Spacers	500,000	16,002,956	25-40	32.01	49.0%
ns 50000 nc 5000	Contigs	5,000	402,606,570	11,391- 149,534	80,521.31	49.0%
ns 50000 nc 5000	Spacers	50,000	1,599,652	25-40	31.99	49.1%
ns 500 nc 5000 HIGH INSERTION RATE	Contigs	5,000	400,779,688	11,241- 149,479	80,155.94	49.0%
ns 500 nc 5000 HIGH INSERTION RATE	Spacers	500	16,007	25-39	32.01	48.9%
ns 75000 nc 5000	Contigs	5,000	401,074,812	10,244- 149,609	80,214.96	49.0%
ns 75000 nc 5000	Spacers	75,000	2,401,592	25-40	32.02	49.0%

1.1.5 Supplementary Table S5: Simulation and job-generation configuration from `Prepare_all_jobs.ipynb`

Dataset group	Dataset(s)	Sequence generation parameters	Tool-run parameters
Real-data subsamples	<code>fraction_0.0005, 0.001, 0.005, 0.01</code>	Subsample IMG/VR4 filtered contigs by class-stratified fraction; full filtered iPHoP spacers	<code>max_distance=5, 38 threads</code>
Real-data subsamples	<code>fraction_0.05, 0.1</code>	Same as above	<code>max_distance=3, 38 threads, skip sassy and indelfree_bruteforce</code>
Full real data	<code>fraction_1</code>	421,431 HQ contigs (class-stratified), full filtered iPHoP spacers	<code>max_distance=3, 38 threads, mem=250G, t=72:00:00, skip sassy and indelfree_bruteforce</code>
Semi-synthetic baseline	<code>ns_3826979_nc_421431_real_base</code>	<code>num_contigs=421431, num_spacers=3826979, contig length 2000-100000, contig GC 46, spacer length 25-50, mismatch range 0-0, spacer insertions 0-0, reverse-complement rate 0.5</code>	<code>max_distance=3, 38 threads, mem=250G, t=72:00:00, skip sassy and indelfree_bruteforce</code>
Fully synthetic (small/medium)	<code>ns_50000_nc_5000, ns_75000_nc_5000, ns_100000_nc_10000, ns_100000_nc_20000, ns_75000_nc_10000</code>	Typical settings: contig distribution normal, spacer distribution normal, spacer GC 49, spacer length 25-40, contig length 10000-150000, spacer insertions 1-5, mismatch range 0-5 (except selected large runs)	Mostly <code>max_distance=5, 38 threads</code>
High-occurrence synthetic	<code>ns_500_nc_5000_HIGH_INSERTION_RATE</code>	SPACER LENGTH 100-2500, spacer length 25-40, contig length 10000-150000, mismatch range 0-5	<code>max_distance=5, 38 threads</code>
Fully synthetic (largest)	<code>ns_500000_nc_100000</code>	contig length up to 550000, mismatch range 0-3	<code>max_distance=3, skip sassy and indelfree_bruteforce</code>

1.1.6 Supplementary Table S6: Recall Values for Each Tool at Different Mismatch Thresholds

Values are counted per spacer-contig alignment (one row per `alignment_idx` in the “multimetric” parquets in zenodo), using exact distance for the stated metric. `n_found` is the number of positive alignments reported by the tool at that exact distance (classification `positive_in_plan` or `positive_not_in_plan`). `n_total` is the total number of positive alignments at that distance in the evaluation set. Recall is `n_found / n_total`.

The `region_idx` field in the source data groups multiple alignments for the same spacer-contig locus; counts here do not collapse by region (as they are in the manuscript).

1.1.6.1 A. IMG/VR4 dataset

Every row lists the results for a specific mismatch threshold and tool. The values are for exact distance (not max), and represent the total number of unique spacer-contig pairs (aligning at that mismatch threshold). The fraction is `tool_matches` divided by `total_possible`.

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
fraction_0.0005	0 edit	blastn	4371	4371	1.000
fraction_0.0005	0 edit	bowtie1	4371	4371	1.000
fraction_0.0005	0 edit	bowtie2	4371	4371	1.000
fraction_0.0005	0 edit	indelfree_bruteforce	4371	4371	1.000
fraction_0.0005	0 edit	indelfree_indexed	4371	4371	1.000
fraction_0.0005	0 edit	minimap2	7	4371	0.002
fraction_0.0005	0 edit	mmseqs2	4287	4371	0.981
fraction_0.0005	0 edit	mummer4	4371	4371	1.000
fraction_0.0005	0 edit	sassy	4371	4371	1.000
fraction_0.0005	0 edit	strobealign	4334	4371	0.992
fraction_0.0005	0 edit	x_mapper	4371	4371	1.000
fraction_0.0005	1 edit	blastn	3213	3491	0.920
fraction_0.0005	1 edit	bowtie1	3456	3491	0.990
fraction_0.0005	1 edit	bowtie2	3491	3491	1.000
fraction_0.0005	1 edit	indelfree_bruteforce	3469	3491	0.994
fraction_0.0005	1 edit	indelfree_indexed	3469	3491	0.994
fraction_0.0005	1 edit	minimap2	0	3491	0.000
fraction_0.0005	1 edit	mmseqs2	3349	3491	0.959
fraction_0.0005	1 edit	mummer4	2618	3491	0.750
fraction_0.0005	1 edit	sassy	3491	3491	1.000
fraction_0.0005	1 edit	strobealign	2882	3491	0.826
fraction_0.0005	1 edit	x_mapper	3147	3491	0.901
fraction_0.0005	2 edit	blastn	2195	3276	0.670
fraction_0.0005	2 edit	bowtie1	3166	3276	0.966
fraction_0.0005	2 edit	bowtie2	3245	3276	0.991
fraction_0.0005	2 edit	indelfree_bruteforce	3204	3276	0.978
fraction_0.0005	2 edit	indelfree_indexed	3204	3276	0.978
fraction_0.0005	2 edit	minimap2	0	3276	0.000
fraction_0.0005	2 edit	mmseqs2	2301	3276	0.702
fraction_0.0005	2 edit	mummer4	1200	3276	0.366
fraction_0.0005	2 edit	sassy	3276	3276	1.000
fraction_0.0005	2 edit	strobealign	1851	3276	0.565
fraction_0.0005	2 edit	x_mapper	2565	3276	0.783
fraction_0.0005	3 edit	blastn	1330	3723	0.357
fraction_0.0005	3 edit	bowtie1	2986	3723	0.802
fraction_0.0005	3 edit	bowtie2	2687	3723	0.722

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
fraction_0.0005	3 edit	indelfree_bruteforce	3058	3723	0.821
fraction_0.0005	3 edit	indelfree_indexed	3058	3723	0.821
fraction_0.0005	3 edit	minimap2	0	3723	0.000
fraction_0.0005	3 edit	mmseqs2	874	3723	0.235
fraction_0.0005	3 edit	mummer4	292	3723	0.078
fraction_0.0005	3 edit	sassy	3723	3723	1.000
fraction_0.0005	3 edit	strobealign	805	3723	0.216
fraction_0.0005	3 edit	x_mapper	1787	3723	0.480
fraction_0.001	0 edit	blastn	6741	6741	1.000
fraction_0.001	0 edit	bowtie1	6741	6741	1.000
fraction_0.001	0 edit	bowtie2	6741	6741	1.000
fraction_0.001	0 edit	indelfree_bruteforce	6741	6741	1.000
fraction_0.001	0 edit	indelfree_indexed	6741	6741	1.000
fraction_0.001	0 edit	minimap2	13	6741	0.002
fraction_0.001	0 edit	mmseqs2	6641	6741	0.985
fraction_0.001	0 edit	mummer4	6741	6741	1.000
fraction_0.001	0 edit	sassy	6741	6741	1.000
fraction_0.001	0 edit	strobealign	6703	6741	0.994
fraction_0.001	0 edit	x_mapper	6741	6741	1.000
fraction_0.001	1 edit	blastn	4724	5112	0.924
fraction_0.001	1 edit	bowtie1	5061	5112	0.990
fraction_0.001	1 edit	bowtie2	5110	5112	1.000
fraction_0.001	1 edit	indelfree_bruteforce	5076	5112	0.993
fraction_0.001	1 edit	indelfree_indexed	5076	5112	0.993
fraction_0.001	1 edit	minimap2	0	5112	0.000
fraction_0.001	1 edit	mmseqs2	4951	5112	0.969
fraction_0.001	1 edit	mummer4	3851	5112	0.753
fraction_0.001	1 edit	sassy	5112	5112	1.000
fraction_0.001	1 edit	strobealign	4442	5112	0.869
fraction_0.001	1 edit	x_mapper	4687	5112	0.917
fraction_0.001	2 edit	blastn	3164	4638	0.682
fraction_0.001	2 edit	bowtie1	4518	4638	0.974
fraction_0.001	2 edit	bowtie2	4614	4638	0.995
fraction_0.001	2 edit	indelfree_bruteforce	4561	4638	0.983
fraction_0.001	2 edit	indelfree_indexed	4561	4638	0.983
fraction_0.001	2 edit	minimap2	0	4638	0.000
fraction_0.001	2 edit	mmseqs2	3491	4638	0.753
fraction_0.001	2 edit	mummer4	1703	4638	0.367
fraction_0.001	2 edit	sassy	4638	4638	1.000
fraction_0.001	2 edit	strobealign	2811	4638	0.606
fraction_0.001	2 edit	x_mapper	3678	4638	0.793
fraction_0.001	3 edit	blastn	2041	5649	0.361
fraction_0.001	3 edit	bowtie1	4581	5649	0.811

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
fraction_0.001	3 edit	bowtie2	4302	5649	0.762
fraction_0.001	3 edit	indelfree_bruteforce	4700	5649	0.832
fraction_0.001	3 edit	indelfree_indexed	4700	5649	0.832
fraction_0.001	3 edit	minimap2	0	5649	0.000
fraction_0.001	3 edit	mmseqs2	1470	5649	0.260
fraction_0.001	3 edit	mummer4	511	5649	0.090
fraction_0.001	3 edit	sassy	5649	5649	1.000
fraction_0.001	3 edit	strobealign	1375	5649	0.243
fraction_0.001	3 edit	x_mapper	2748	5649	0.486
fraction_0.005	0 edit	blastn	21987	21987	1.000
fraction_0.005	0 edit	bowtie1	21987	21987	1.000
fraction_0.005	0 edit	bowtie2	21987	21987	1.000
fraction_0.005	0 edit	indelfree_indexed	21987	21987	1.000
fraction_0.005	0 edit	minimap2	52	21987	0.002
fraction_0.005	0 edit	mmseqs2	21687	21987	0.986
fraction_0.005	0 edit	mummer4	21987	21987	1.000
fraction_0.005	0 edit	sassy	21987	21987	1.000
fraction_0.005	0 edit	strobealign	21754	21987	0.989
fraction_0.005	0 edit	x_mapper	21978	21987	1.000
fraction_0.005	1 edit	blastn	16349	17580	0.930
fraction_0.005	1 edit	bowtie1	17375	17580	0.988
fraction_0.005	1 edit	bowtie2	17576	17580	1.000
fraction_0.005	1 edit	indelfree_indexed	17455	17580	0.993
fraction_0.005	1 edit	minimap2	0	17580	0.000
fraction_0.005	1 edit	mmseqs2	16976	17580	0.966
fraction_0.005	1 edit	mummer4	12816	17580	0.729
fraction_0.005	1 edit	sassy	17580	17580	1.000
fraction_0.005	1 edit	strobealign	14843	17580	0.844
fraction_0.005	1 edit	x_mapper	14423	17580	0.820
fraction_0.005	2 edit	blastn	11439	16648	0.687
fraction_0.005	2 edit	bowtie1	16144	16648	0.970
fraction_0.005	2 edit	bowtie2	16504	16648	0.991
fraction_0.005	2 edit	indelfree_indexed	16321	16648	0.980
fraction_0.005	2 edit	minimap2	0	16648	0.000
fraction_0.005	2 edit	mmseqs2	11948	16648	0.718
fraction_0.005	2 edit	mummer4	6067	16648	0.364
fraction_0.005	2 edit	sassy	16648	16648	1.000
fraction_0.005	2 edit	strobealign	9428	16648	0.566
fraction_0.005	2 edit	x_mapper	10297	16648	0.619
fraction_0.005	3 edit	blastn	8114	23556	0.344
fraction_0.005	3 edit	bowtie1	18395	23556	0.781
fraction_0.005	3 edit	bowtie2	16912	23556	0.718
fraction_0.005	3 edit	indelfree_indexed	18919	23556	0.803

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
fraction_0.005	3 edit	minimap2	0	23556	0.000
fraction_0.005	3 edit	mmseqs2	5600	23556	0.238
fraction_0.005	3 edit	mummer4	1884	23556	0.080
fraction_0.005	3 edit	sassy	23556	23556	1.000
fraction_0.005	3 edit	strobealign	5065	23556	0.215
fraction_0.005	3 edit	x_mapper	6963	23556	0.296
fraction_0.01	0 edit	blastn	53076	53076	1.000
fraction_0.01	0 edit	bowtie1	53076	53076	1.000
fraction_0.01	0 edit	bowtie2	53076	53076	1.000
fraction_0.01	0 edit	indelfree_indexed	53076	53076	1.000
fraction_0.01	0 edit	minimap2	73	53076	0.001
fraction_0.01	0 edit	mmseqs2	52496	53076	0.989
fraction_0.01	0 edit	mummer4	53076	53076	1.000
fraction_0.01	0 edit	sassy	53076	53076	1.000
fraction_0.01	0 edit	strobealign	52440	53076	0.988
fraction_0.01	0 edit	x_mapper	53033	53076	0.999
fraction_0.01	1 edit	blastn	37727	40519	0.931
fraction_0.01	1 edit	bowtie1	40184	40519	0.992
fraction_0.01	1 edit	bowtie2	40517	40519	1.000
fraction_0.01	1 edit	indelfree_indexed	40257	40519	0.994
fraction_0.01	1 edit	minimap2	0	40519	0.000
fraction_0.01	1 edit	mmseqs2	39256	40519	0.969
fraction_0.01	1 edit	mummer4	29741	40519	0.734
fraction_0.01	1 edit	sassy	40519	40519	1.000
fraction_0.01	1 edit	strobealign	32629	40519	0.805
fraction_0.01	1 edit	x_mapper	30931	40519	0.763
fraction_0.01	2 edit	blastn	26325	38352	0.686
fraction_0.01	2 edit	bowtie1	37321	38352	0.973
fraction_0.01	2 edit	bowtie2	38102	38352	0.993
fraction_0.01	2 edit	indelfree_indexed	37705	38352	0.983
fraction_0.01	2 edit	minimap2	0	38352	0.000
fraction_0.01	2 edit	mmseqs2	27735	38352	0.723
fraction_0.01	2 edit	mummer4	13529	38352	0.353
fraction_0.01	2 edit	sassy	38352	38352	1.000
fraction_0.01	2 edit	strobealign	20240	38352	0.528
fraction_0.01	2 edit	x_mapper	20165	38352	0.526
fraction_0.01	3 edit	blastn	18347	52946	0.347
fraction_0.01	3 edit	bowtie1	41223	52946	0.779
fraction_0.01	3 edit	bowtie2	38152	52946	0.721
fraction_0.01	3 edit	indelfree_indexed	42364	52946	0.800
fraction_0.01	3 edit	minimap2	0	52946	0.000
fraction_0.01	3 edit	mmseqs2	12717	52946	0.240
fraction_0.01	3 edit	mummer4	4157	52946	0.079

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
fraction_0.01	3 edit	sassy	52946	52946	1.000
fraction_0.01	3 edit	strobealign	10441	52946	0.197
fraction_0.01	3 edit	x_mapper	12491	52946	0.236
fraction_1	0 hamming	blastn	7486741	7486741	1.000
fraction_1	0 hamming	bowtie1	7486741	7486741	1.000
fraction_1	0 hamming	bowtie2	7486741	7486741	1.000
fraction_1	0 hamming	minimap2	7509	7486741	0.001
fraction_1	0 hamming	mmseqs2	7300313	7486741	0.975
fraction_1	0 hamming	mummer4	7486741	7486741	1.000
fraction_1	0 hamming	strobealign	5893368	7486741	0.787
fraction_1	0 hamming	x_mapper	4087837	7486741	0.546
fraction_1	1 hamming	blastn	4300028	5349840	0.804
fraction_1	1 hamming	bowtie1	5348252	5349840	1.000
fraction_1	1 hamming	bowtie2	5348360	5349840	1.000
fraction_1	1 hamming	minimap2	0	5349840	0.000
fraction_1	1 hamming	mmseqs2	4576693	5349840	0.855
fraction_1	1 hamming	mummer4	3574831	5349840	0.668
fraction_1	1 hamming	strobealign	1865559	5349840	0.349
fraction_1	1 hamming	x_mapper	1382061	5349840	0.258
fraction_1	2 hamming	blastn	3295366	5191692	0.635
fraction_1	2 hamming	bowtie1	4840011	5191692	0.932
fraction_1	2 hamming	bowtie2	4827887	5191692	0.930
fraction_1	2 hamming	minimap2	0	5191692	0.000
fraction_1	2 hamming	mmseqs2	3160802	5191692	0.609
fraction_1	2 hamming	mummer4	1819317	5191692	0.350
fraction_1	2 hamming	strobealign	610239	5191692	0.118
fraction_1	2 hamming	x_mapper	503854	5191692	0.097
fraction_1	3 hamming	blastn	3253234	6584617	0.494
fraction_1	3 hamming	bowtie1	5640628	6584617	0.857
fraction_1	3 hamming	bowtie2	4996518	6584617	0.759
fraction_1	3 hamming	minimap2	0	6584617	0.000
fraction_1	3 hamming	mmseqs2	2072104	6584617	0.315
fraction_1	3 hamming	mummer4	778217	6584617	0.118
fraction_1	3 hamming	strobealign	181230	6584617	0.028
fraction_1	3 hamming	x_mapper	134350	6584617	0.020

1.1.6.2 B. Synthetic dataset

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
ns_100000_nc_10000	0 edit	blastn	50393	50393	1.000
ns_100000_nc_10000	0 edit	bowtie1	50393	50393	1.000
ns_100000_nc_10000	0 edit	bowtie2	50393	50393	1.000
ns_100000_nc_10000	0 edit	indelfree_bruteforce	50393	50393	1.000

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
ns_100000_nc_10000	0 edit	indelfree_indexed	50393	50393	1.000
ns_100000_nc_10000	0 edit	minimap2	45	50393	0.001
ns_100000_nc_10000	0 edit	mmseqs2	50237	50393	0.997
ns_100000_nc_10000	0 edit	mummer4	50393	50393	1.000
ns_100000_nc_10000	0 edit	sassy	50393	50393	1.000
ns_100000_nc_10000	0 edit	strobealign	49732	50393	0.987
ns_100000_nc_10000	0 edit	x_mapper	50159	50393	0.995
ns_100000_nc_10000	1 edit	blastn	40655	54891	0.741
ns_100000_nc_10000	1 edit	bowtie1	50199	54891	0.915
ns_100000_nc_10000	1 edit	bowtie2	50210	54891	0.915
ns_100000_nc_10000	1 edit	indelfree_bruteforce	50199	54891	0.915
ns_100000_nc_10000	1 edit	indelfree_indexed	50199	54891	0.915
ns_100000_nc_10000	1 edit	mmseqs2	44806	54891	0.816
ns_100000_nc_10000	1 edit	mummer4	31028	54891	0.565
ns_100000_nc_10000	1 edit	sassy	50325	54891	0.917
ns_100000_nc_10000	1 edit	strobealign	37368	54891	0.681
ns_100000_nc_10000	1 edit	x_mapper	24769	54891	0.451
ns_100000_nc_10000	2 edit	blastn	34597	63663	0.543
ns_100000_nc_10000	2 edit	bowtie1	49873	63663	0.783
ns_100000_nc_10000	2 edit	bowtie2	49857	63663	0.783
ns_100000_nc_10000	2 edit	indelfree_bruteforce	49875	63663	0.783
ns_100000_nc_10000	2 edit	indelfree_indexed	49875	63663	0.783
ns_100000_nc_10000	2 edit	mmseqs2	32941	63663	0.517
ns_100000_nc_10000	2 edit	mummer4	14945	63663	0.235
ns_100000_nc_10000	2 edit	sassy	50180	63663	0.788
ns_100000_nc_10000	2 edit	strobealign	21326	63663	0.335
ns_100000_nc_10000	2 edit	x_mapper	10942	63663	0.172
ns_100000_nc_10000	3 edit	blastn	30879	74834	0.413
ns_100000_nc_10000	3 edit	bowtie1	49653	74834	0.664
ns_100000_nc_10000	3 edit	bowtie2	45156	74834	0.603
ns_100000_nc_10000	3 edit	indelfree_bruteforce	49993	74834	0.668
ns_100000_nc_10000	3 edit	indelfree_indexed	49993	74834	0.668
ns_100000_nc_10000	3 edit	mmseqs2	18510	74834	0.247
ns_100000_nc_10000	3 edit	mummer4	5855	74834	0.078
ns_100000_nc_10000	3 edit	sassy	51274	74834	0.685
ns_100000_nc_10000	3 edit	strobealign	8823	74834	0.118
ns_100000_nc_10000	3 edit	x_mapper	3923	74834	0.052
ns_100000_nc_10000	4 edit	blastn	29083	102345	0.284
ns_100000_nc_10000	4 edit	bowtie2	219	102345	0.002
ns_100000_nc_10000	4 edit	indelfree_bruteforce	52114	102345	0.509
ns_100000_nc_10000	4 edit	indelfree_indexed	52101	102345	0.509
ns_100000_nc_10000	4 edit	mmseqs2	9456	102345	0.092
ns_100000_nc_10000	4 edit	mummer4	434	102345	0.004

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
ns_100000_nc_10000	4 edit	sassy	71853	102345	0.702
ns_100000_nc_10000	4 edit	strobealign	2042	102345	0.020
ns_100000_nc_10000	4 edit	x_mapper	1252	102345	0.012
ns_100000_nc_10000	5 edit	blastn	23827	544086	0.044
ns_100000_nc_10000	5 edit	indelfree_bruteforce	68700	544086	0.126
ns_100000_nc_10000	5 edit	indelfree_indexed	68027	544086	0.125
ns_100000_nc_10000	5 edit	mmseqs2	3610	544086	0.007
ns_100000_nc_10000	5 edit	mummer4	13	544086	0.000
ns_100000_nc_10000	5 edit	sassy	511134	544086	0.939
ns_100000_nc_10000	5 edit	strobealign	155	544086	0.000
ns_100000_nc_10000	5 edit	x_mapper	460	544086	0.001
ns_100000_nc_10000	0 hamming	blastn	50393	50393	1.000
ns_100000_nc_10000	0 hamming	bowtie1	50393	50393	1.000
ns_100000_nc_10000	0 hamming	bowtie2	50393	50393	1.000
ns_100000_nc_10000	0 hamming	indelfree_bruteforce	50393	50393	1.000
ns_100000_nc_10000	0 hamming	indelfree_indexed	50393	50393	1.000
ns_100000_nc_10000	0 hamming	minimap2	45	50393	0.001
ns_100000_nc_10000	0 hamming	mmseqs2	50237	50393	0.997
ns_100000_nc_10000	0 hamming	mummer4	50393	50393	1.000
ns_100000_nc_10000	0 hamming	sassy	50393	50393	1.000
ns_100000_nc_10000	0 hamming	strobealign	49732	50393	0.987
ns_100000_nc_10000	0 hamming	x_mapper	50159	50393	0.995
ns_100000_nc_10000	1 hamming	blastn	40655	53380	0.762
ns_100000_nc_10000	1 hamming	bowtie1	50199	53380	0.940
ns_100000_nc_10000	1 hamming	bowtie2	50199	53380	0.940
ns_100000_nc_10000	1 hamming	indelfree_bruteforce	50199	53380	0.940
ns_100000_nc_10000	1 hamming	indelfree_indexed	50199	53380	0.940
ns_100000_nc_10000	1 hamming	mmseqs2	44396	53380	0.832
ns_100000_nc_10000	1 hamming	mummer4	31011	53380	0.581
ns_100000_nc_10000	1 hamming	sassy	49184	53380	0.921
ns_100000_nc_10000	1 hamming	strobealign	37367	53380	0.700
ns_100000_nc_10000	1 hamming	x_mapper	24767	53380	0.464
ns_100000_nc_10000	2 hamming	blastn	34448	60981	0.565
ns_100000_nc_10000	2 hamming	bowtie1	49809	60981	0.817
ns_100000_nc_10000	2 hamming	bowtie2	49692	60981	0.815
ns_100000_nc_10000	2 hamming	indelfree_bruteforce	49809	60981	0.817
ns_100000_nc_10000	2 hamming	indelfree_indexed	49809	60981	0.817
ns_100000_nc_10000	2 hamming	mmseqs2	31822	60981	0.522
ns_100000_nc_10000	2 hamming	mummer4	14786	60981	0.242
ns_100000_nc_10000	2 hamming	sassy	48720	60981	0.799
ns_100000_nc_10000	2 hamming	strobealign	21316	60981	0.350
ns_100000_nc_10000	2 hamming	x_mapper	10923	60981	0.179
ns_100000_nc_10000	3 hamming	blastn	30414	71296	0.427

Fraction/Simulation Distance		Metric	Tool	n_found	n_total	Recall
ns_100000_nc_10000	3	hamming	bowtie1	49717	71296	0.697
ns_100000_nc_10000	3	hamming	bowtie2	45107	71296	0.633
ns_100000_nc_10000	3	hamming	indelfree_bruteforce	49717	71296	0.697
ns_100000_nc_10000	3	hamming	indelfree_indexed	49717	71296	0.697
ns_100000_nc_10000	3	hamming	mmseqs2	18129	71296	0.254
ns_100000_nc_10000	3	hamming	mummer4	5741	71296	0.081
ns_100000_nc_10000	3	hamming	sassy	48630	71296	0.682
ns_100000_nc_10000	3	hamming	strobealign	8822	71296	0.124
ns_100000_nc_10000	3	hamming	x_mapper	3908	71296	0.055
ns_100000_nc_10000	4	hamming	blastn	27950	79017	0.354
ns_100000_nc_10000	4	hamming	bowtie2	331	79017	0.004
ns_100000_nc_10000	4	hamming	indelfree_bruteforce	51389	79017	0.650
ns_100000_nc_10000	4	hamming	indelfree_indexed	51380	79017	0.650
ns_100000_nc_10000	4	hamming	mmseqs2	9501	79017	0.120
ns_100000_nc_10000	4	hamming	mummer4	463	79017	0.006
ns_100000_nc_10000	4	hamming	sassy	49795	79017	0.630
ns_100000_nc_10000	4	hamming	strobealign	2049	79017	0.026
ns_100000_nc_10000	4	hamming	x_mapper	1254	79017	0.016
ns_100000_nc_10000	5	hamming	blastn	21489	97256	0.221
ns_100000_nc_10000	5	hamming	bowtie2	95	97256	0.001
ns_100000_nc_10000	5	hamming	indelfree_bruteforce	69767	97256	0.717
ns_100000_nc_10000	5	hamming	indelfree_indexed	69090	97256	0.710
ns_100000_nc_10000	5	hamming	mmseqs2	4238	97256	0.044
ns_100000_nc_10000	5	hamming	mummer4	23	97256	0.000
ns_100000_nc_10000	5	hamming	sassy	65899	97256	0.678
ns_100000_nc_10000	5	hamming	strobealign	159	97256	0.002
ns_100000_nc_10000	5	hamming	x_mapper	475	97256	0.005
ns_100000_nc_20000	0	edit	blastn	74758	74758	1.000
ns_100000_nc_20000	0	edit	bowtie1	74758	74758	1.000
ns_100000_nc_20000	0	edit	bowtie2	74758	74758	1.000
ns_100000_nc_20000	0	edit	indelfree_bruteforce	74758	74758	1.000
ns_100000_nc_20000	0	edit	indelfree_indexed	74758	74758	1.000
ns_100000_nc_20000	0	edit	minimap2	92	74758	0.001
ns_100000_nc_20000	0	edit	mmseqs2	74519	74758	0.997
ns_100000_nc_20000	0	edit	mummer4	74758	74758	1.000
ns_100000_nc_20000	0	edit	sassy	74758	74758	1.000
ns_100000_nc_20000	0	edit	strobealign	73490	74758	0.983
ns_100000_nc_20000	0	edit	x_mapper	73993	74758	0.990
ns_100000_nc_20000	1	edit	blastn	61186	82512	0.742
ns_100000_nc_20000	1	edit	bowtie1	75347	82512	0.913
ns_100000_nc_20000	1	edit	bowtie2	75371	82512	0.913
ns_100000_nc_20000	1	edit	indelfree_bruteforce	75347	82512	0.913
ns_100000_nc_20000	1	edit	indelfree_indexed	75347	82512	0.913

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
ns_100000_nc_20000	1 edit	mmseqs2	67112	82512	0.813
ns_100000_nc_20000	1 edit	mummer4	46259	82512	0.561
ns_100000_nc_20000	1 edit	sassy	75531	82512	0.915
ns_100000_nc_20000	1 edit	strobealign	51532	82512	0.625
ns_100000_nc_20000	1 edit	x_mapper	32941	82512	0.399
ns_100000_nc_20000	2 edit	blastn	51583	95366	0.541
ns_100000_nc_20000	2 edit	bowtie1	74711	95366	0.783
ns_100000_nc_20000	2 edit	bowtie2	74623	95366	0.782
ns_100000_nc_20000	2 edit	indelfree_bruteforce	74711	95366	0.783
ns_100000_nc_20000	2 edit	indelfree_indexed	74711	95366	0.783
ns_100000_nc_20000	2 edit	mmseqs2	49064	95366	0.514
ns_100000_nc_20000	2 edit	mummer4	22594	95366	0.237
ns_100000_nc_20000	2 edit	sassy	75144	95366	0.788
ns_100000_nc_20000	2 edit	strobealign	27883	95366	0.292
ns_100000_nc_20000	2 edit	x_mapper	13472	95366	0.141
ns_100000_nc_20000	3 edit	blastn	46655	112339	0.415
ns_100000_nc_20000	3 edit	bowtie1	75432	112339	0.671
ns_100000_nc_20000	3 edit	bowtie2	68585	112339	0.611
ns_100000_nc_20000	3 edit	indelfree_bruteforce	75464	112339	0.672
ns_100000_nc_20000	3 edit	indelfree_indexed	75464	112339	0.672
ns_100000_nc_20000	3 edit	mmseqs2	28067	112339	0.250
ns_100000_nc_20000	3 edit	mummer4	9070	112339	0.081
ns_100000_nc_20000	3 edit	sassy	76396	112339	0.680
ns_100000_nc_20000	3 edit	strobealign	11589	112339	0.103
ns_100000_nc_20000	3 edit	x_mapper	3586	112339	0.032
ns_100000_nc_20000	4 edit	blastn	21389	69625	0.307
ns_100000_nc_20000	4 edit	indelfree_bruteforce	3621	69625	0.052
ns_100000_nc_20000	4 edit	indelfree_indexed	3609	69625	0.052
ns_100000_nc_20000	4 edit	mmseqs2	8921	69625	0.128
ns_100000_nc_20000	4 edit	mummer4	3	69625	0.000
ns_100000_nc_20000	4 edit	sassy	41142	69625	0.591
ns_100000_nc_20000	4 edit	strobealign	1	69625	0.000
ns_100000_nc_20000	4 edit	x_mapper	13	69625	0.000
ns_100000_nc_20000	5 edit	blastn	18273	941140	0.019
ns_100000_nc_20000	5 edit	indelfree_bruteforce	38318	941140	0.041
ns_100000_nc_20000	5 edit	indelfree_indexed	37153	941140	0.039
ns_100000_nc_20000	5 edit	mmseqs2	2404	941140	0.003
ns_100000_nc_20000	5 edit	sassy	914758	941140	0.972
ns_100000_nc_20000	5 edit	x_mapper	28	941140	0.000
ns_100000_nc_20000	0 hamming	blastn	74758	74758	1.000
ns_100000_nc_20000	0 hamming	bowtie1	74758	74758	1.000
ns_100000_nc_20000	0 hamming	bowtie2	74758	74758	1.000
ns_100000_nc_20000	0 hamming	indelfree_bruteforce	74758	74758	1.000

Fraction/Simulation Distance		Metric	Tool	n_found	n_total	Recall
ns_100000_nc_20000	0	hamming	indelfree_indexed	74758	74758	1.000
ns_100000_nc_20000	0	hamming	minimap2	92	74758	0.001
ns_100000_nc_20000	0	hamming	mmseqs2	74519	74758	0.997
ns_100000_nc_20000	0	hamming	mummer4	74758	74758	1.000
ns_100000_nc_20000	0	hamming	sassy	74758	74758	1.000
ns_100000_nc_20000	0	hamming	strobealign	73490	74758	0.983
ns_100000_nc_20000	0	hamming	x_mapper	73993	74758	0.990
ns_100000_nc_20000	1	hamming	blastn	61186	80116	0.764
ns_100000_nc_20000	1	hamming	bowtie1	75347	80116	0.940
ns_100000_nc_20000	1	hamming	bowtie2	75347	80116	0.940
ns_100000_nc_20000	1	hamming	indelfree_bruteforce	75347	80116	0.940
ns_100000_nc_20000	1	hamming	indelfree_indexed	75347	80116	0.940
ns_100000_nc_20000	1	hamming	mmseqs2	66470	80116	0.830
ns_100000_nc_20000	1	hamming	mummer4	46209	80116	0.577
ns_100000_nc_20000	1	hamming	sassy	73723	80116	0.920
ns_100000_nc_20000	1	hamming	strobealign	51532	80116	0.643
ns_100000_nc_20000	1	hamming	x_mapper	32937	80116	0.411
ns_100000_nc_20000	2	hamming	blastn	51400	91610	0.561
ns_100000_nc_20000	2	hamming	bowtie1	74612	91610	0.814
ns_100000_nc_20000	2	hamming	bowtie2	74438	91610	0.813
ns_100000_nc_20000	2	hamming	indelfree_bruteforce	74612	91610	0.814
ns_100000_nc_20000	2	hamming	indelfree_indexed	74612	91610	0.814
ns_100000_nc_20000	2	hamming	mmseqs2	47530	91610	0.519
ns_100000_nc_20000	2	hamming	mummer4	22407	91610	0.245
ns_100000_nc_20000	2	hamming	sassy	73034	91610	0.797
ns_100000_nc_20000	2	hamming	strobealign	27869	91610	0.304
ns_100000_nc_20000	2	hamming	x_mapper	13455	91610	0.147
ns_100000_nc_20000	3	hamming	blastn	46061	108401	0.425
ns_100000_nc_20000	3	hamming	bowtie1	75531	108401	0.697
ns_100000_nc_20000	3	hamming	bowtie2	68620	108401	0.633
ns_100000_nc_20000	3	hamming	indelfree_bruteforce	75531	108401	0.697
ns_100000_nc_20000	3	hamming	indelfree_indexed	75531	108401	0.697
ns_100000_nc_20000	3	hamming	mmseqs2	27654	108401	0.255
ns_100000_nc_20000	3	hamming	mummer4	8962	108401	0.083
ns_100000_nc_20000	3	hamming	sassy	73647	108401	0.679
ns_100000_nc_20000	3	hamming	strobealign	11603	108401	0.107
ns_100000_nc_20000	3	hamming	x_mapper	3598	108401	0.033
ns_100000_nc_20000	4	hamming	blastn	20136	35095	0.574
ns_100000_nc_20000	4	hamming	bowtie2	109	35095	0.003
ns_100000_nc_20000	4	hamming	indelfree_bruteforce	2910	35095	0.083
ns_100000_nc_20000	4	hamming	indelfree_indexed	2905	35095	0.083
ns_100000_nc_20000	4	hamming	mmseqs2	9283	35095	0.265
ns_100000_nc_20000	4	hamming	mummer4	26	35095	0.001

Fraction/Simulation Distance		Metric	Tool	n_found	n_total	Recall
ns_100000_nc_20000	4	hamming	sassy	7616	35095	0.217
ns_100000_nc_20000	4	hamming	strobealign	1	35095	0.000
ns_100000_nc_20000	4	hamming	x_mapper	11	35095	0.000
ns_100000_nc_20000	5	hamming	blastn	16256	59330	0.274
ns_100000_nc_20000	5	hamming	bowtie2	48	59330	0.001
ns_100000_nc_20000	5	hamming	indelfree_bruteforce	39061	59330	0.658
ns_100000_nc_20000	5	hamming	indelfree_indexed	37889	59330	0.639
ns_100000_nc_20000	5	hamming	mmseqs2	3121	59330	0.053
ns_100000_nc_20000	5	hamming	mummer4	3	59330	0.000
ns_100000_nc_20000	5	hamming	sassy	34112	59330	0.575
ns_100000_nc_20000	5	hamming	x_mapper	28	59330	0.000
ns_100_nc_5000	0	edit	blastn	30	30	1.000
ns_100_nc_5000	0	edit	bowtie1	30	30	1.000
ns_100_nc_5000	0	edit	bowtie2	30	30	1.000
ns_100_nc_5000	0	edit	indelfree_bruteforce	30	30	1.000
ns_100_nc_5000	0	edit	indelfree_indexed	30	30	1.000
ns_100_nc_5000	0	edit	mmseqs2	30	30	1.000
ns_100_nc_5000	0	edit	sassy	30	30	1.000
ns_100_nc_5000	0	edit	x_mapper	30	30	1.000
ns_100_nc_5000	1	edit	blastn	20	22	0.909
ns_100_nc_5000	1	edit	bowtie1	22	22	1.000
ns_100_nc_5000	1	edit	bowtie2	22	22	1.000
ns_100_nc_5000	1	edit	indelfree_bruteforce	22	22	1.000
ns_100_nc_5000	1	edit	indelfree_indexed	22	22	1.000
ns_100_nc_5000	1	edit	mmseqs2	20	22	0.909
ns_100_nc_5000	1	edit	sassy	22	22	1.000
ns_100_nc_5000	1	edit	x_mapper	12	22	0.545
ns_100_nc_5000	2	edit	blastn	36	56	0.643
ns_100_nc_5000	2	edit	bowtie1	45	56	0.804
ns_100_nc_5000	2	edit	bowtie2	45	56	0.804
ns_100_nc_5000	2	edit	indelfree_bruteforce	45	56	0.804
ns_100_nc_5000	2	edit	indelfree_indexed	45	56	0.804
ns_100_nc_5000	2	edit	mmseqs2	31	56	0.554
ns_100_nc_5000	2	edit	sassy	45	56	0.804
ns_100_nc_5000	2	edit	x_mapper	12	56	0.214
ns_100_nc_5000	3	edit	blastn	20	48	0.417
ns_100_nc_5000	3	edit	bowtie1	33	48	0.688
ns_100_nc_5000	3	edit	bowtie2	24	48	0.500
ns_100_nc_5000	3	edit	indelfree_bruteforce	33	48	0.688
ns_100_nc_5000	3	edit	indelfree_indexed	33	48	0.688
ns_100_nc_5000	3	edit	mmseqs2	14	48	0.292
ns_100_nc_5000	3	edit	sassy	34	48	0.708
ns_100_nc_5000	3	edit	x_mapper	7	48	0.146

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
ns_100_nc_5000	4 edit	blastn	11	88	0.125
ns_100_nc_5000	4 edit	bowtie2	7	88	0.080
ns_100_nc_5000	4 edit	indelfree_bruteforce	31	88	0.352
ns_100_nc_5000	4 edit	indelfree_indexed	31	88	0.352
ns_100_nc_5000	4 edit	mmseqs2	10	88	0.114
ns_100_nc_5000	4 edit	sassy	76	88	0.864
ns_100_nc_5000	4 edit	x_mapper	3	88	0.034
ns_100_nc_5000	5 edit	blastn	22	1048	0.021
ns_100_nc_5000	5 edit	indelfree_bruteforce	77	1048	0.073
ns_100_nc_5000	5 edit	indelfree_indexed	75	1048	0.072
ns_100_nc_5000	5 edit	mmseqs2	5	1048	0.005
ns_100_nc_5000	5 edit	sassy	1016	1048	0.969
ns_100_nc_5000	0 hamming	blastn	30	30	1.000
ns_100_nc_5000	0 hamming	bowtie1	30	30	1.000
ns_100_nc_5000	0 hamming	bowtie2	30	30	1.000
ns_100_nc_5000	0 hamming	indelfree_bruteforce	30	30	1.000
ns_100_nc_5000	0 hamming	indelfree_indexed	30	30	1.000
ns_100_nc_5000	0 hamming	mmseqs2	30	30	1.000
ns_100_nc_5000	0 hamming	sassy	30	30	1.000
ns_100_nc_5000	0 hamming	x_mapper	30	30	1.000
ns_100_nc_5000	1 hamming	blastn	20	22	0.909
ns_100_nc_5000	1 hamming	bowtie1	22	22	1.000
ns_100_nc_5000	1 hamming	bowtie2	22	22	1.000
ns_100_nc_5000	1 hamming	indelfree_bruteforce	22	22	1.000
ns_100_nc_5000	1 hamming	indelfree_indexed	22	22	1.000
ns_100_nc_5000	1 hamming	mmseqs2	20	22	0.909
ns_100_nc_5000	1 hamming	sassy	22	22	1.000
ns_100_nc_5000	1 hamming	x_mapper	12	22	0.545
ns_100_nc_5000	2 hamming	blastn	36	52	0.692
ns_100_nc_5000	2 hamming	bowtie1	45	52	0.865
ns_100_nc_5000	2 hamming	bowtie2	45	52	0.865
ns_100_nc_5000	2 hamming	indelfree_bruteforce	45	52	0.865
ns_100_nc_5000	2 hamming	indelfree_indexed	45	52	0.865
ns_100_nc_5000	2 hamming	mmseqs2	29	52	0.558
ns_100_nc_5000	2 hamming	sassy	43	52	0.827
ns_100_nc_5000	2 hamming	x_mapper	12	52	0.231
ns_100_nc_5000	3 hamming	blastn	20	47	0.426
ns_100_nc_5000	3 hamming	bowtie1	33	47	0.702
ns_100_nc_5000	3 hamming	bowtie2	24	47	0.511
ns_100_nc_5000	3 hamming	indelfree_bruteforce	33	47	0.702
ns_100_nc_5000	3 hamming	indelfree_indexed	33	47	0.702
ns_100_nc_5000	3 hamming	mmseqs2	16	47	0.340
ns_100_nc_5000	3 hamming	sassy	31	47	0.660

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
ns_100_nc_5000	3 hamming	x_mapper	7	47	0.149
ns_100_nc_5000	4 hamming	blastn	11	41	0.268
ns_100_nc_5000	4 hamming	bowtie2	7	41	0.171
ns_100_nc_5000	4 hamming	indelfree_bruteforce	29	41	0.707
ns_100_nc_5000	4 hamming	indelfree_indexed	29	41	0.707
ns_100_nc_5000	4 hamming	mmseqs2	8	41	0.195
ns_100_nc_5000	4 hamming	sassy	31	41	0.756
ns_100_nc_5000	4 hamming	x_mapper	3	41	0.073
ns_100_nc_5000	5 hamming	blastn	20	100	0.200
ns_100_nc_5000	5 hamming	indelfree_bruteforce	79	100	0.790
ns_100_nc_5000	5 hamming	indelfree_indexed	77	100	0.770
ns_100_nc_5000	5 hamming	mmseqs2	7	100	0.070
ns_100_nc_5000	5 hamming	sassy	68	100	0.680
ns_3826979_nc_421431_real_0.1	blastn	blastn	1	1	1.000
ns_3826979_nc_421431_real_0.1	blastn	blastn	1	1	1.000
ns_3826979_nc_421431_real_0.1	blastn	blastn	1	1	1.000
ns_3826979_nc_421431_real_0.1	blastn	blastn	1	1	1.000
ns_3826979_nc_421431_real_0.1	blastn	blastn	1	1	1.000
ns_3826979_nc_421431_real_0.1	blastn	blastn	1	1	1.000
ns_3826979_nc_421431_real_0.1	blastn	blastn	11	101	0.109
ns_3826979_nc_421431_real_0.1	blastn	blastn	46	101	0.455
ns_3826979_nc_421431_real_0.1	blastn	blastn	97	101	0.960
ns_3826979_nc_421431_real_0.1	blastn	blastn	31	101	0.307
ns_3826979_nc_421431_real_0.1	blastn	blastn	10	101	0.099
ns_3826979_nc_421431_real_0.1	blastn	blastn	28	101	0.277
ns_3826979_nc_421431_real_0.1	blastn	blastn	6	101	0.059
ns_3826979_nc_421431_real_0.1	blastn	blastn	9	101	0.089
ns_3826979_nc_421431_real_0.2	blastn	blastn	324	6135	0.053
ns_3826979_nc_421431_real_0.2	blastn	blastn	2293	6135	0.374
ns_3826979_nc_421431_real_0.2	blastn	blastn	5800	6135	0.945
ns_3826979_nc_421431_real_0.2	blastn	blastn	1434	6135	0.234
ns_3826979_nc_421431_real_0.2	blastn	blastn	85	6135	0.014
ns_3826979_nc_421431_real_0.2	blastn	blastn	185	6135	0.030
ns_3826979_nc_421431_real_0.2	blastn	blastn	60	6135	0.010
ns_3826979_nc_421431_real_0.2	blastn	blastn	163	6135	0.027
ns_3826979_nc_421431_real_0.3	blastn	blastn	3212	55965	0.057
ns_3826979_nc_421431_real_0.3	blastn	blastn	52048	55965	0.930
ns_3826979_nc_421431_real_0.3	blastn	blastn	6805	55965	0.122
ns_3826979_nc_421431_real_0.3	blastn	blastn	32326	55965	0.578
ns_3826979_nc_421431_real_0.3	blastn	blastn	202	55965	0.004
ns_3826979_nc_421431_real_0.3	blastn	blastn	377	55965	0.007
ns_3826979_nc_421431_real_0.3	blastn	blastn	104	55965	0.002
ns_3826979_nc_421431_real_0.3	blastn	blastn	1134	55965	0.020

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
ns_3826979_nc_421431_real	baseline	blastn	13976	19579	0.714
ns_3826979_nc_421431_real	baseline	bowtie2	1	19579	0.000
ns_3826979_nc_421431_real	baseline	indelfree_indexed	4	19579	0.000
ns_3826979_nc_421431_real	baseline	mmseqs2	215	19579	0.011
ns_3826979_nc_421431_real	baseline	mummer4	52	19579	0.003
ns_3826979_nc_421431_real	baseline	strobealign	14	19579	0.001
ns_3826979_nc_421431_real	baseline	x_mapper	5336	19579	0.273
ns_3826979_nc_421431_real	baseline	blastn	45644	59900	0.762
ns_3826979_nc_421431_real	baseline	mmseqs2	169	59900	0.003
ns_3826979_nc_421431_real	baseline	mummer4	1	59900	0.000
ns_3826979_nc_421431_real	baseline	strobealign	3	59900	0.000
ns_3826979_nc_421431_real	baseline	x_mapper	14106	59900	0.235
ns_3826979_nc_421431_real	baseline	blastn	1	1	1.000
ns_3826979_nc_421431_real	baseline	bowtie1	1	1	1.000
ns_3826979_nc_421431_real	baseline	bowtie2	1	1	1.000
ns_3826979_nc_421431_real	baseline	indelfree_indexed	1	1	1.000
ns_3826979_nc_421431_real	baseline	mmseqs2	1	1	1.000
ns_3826979_nc_421431_real	baseline	mummer4	1	1	1.000
ns_3826979_nc_421431_real	baseline	blastn	11	48	0.229
ns_3826979_nc_421431_real	baseline	bowtie1	46	48	0.958
ns_3826979_nc_421431_real	baseline	bowtie2	46	48	0.958
ns_3826979_nc_421431_real	baseline	indelfree_indexed	31	48	0.646
ns_3826979_nc_421431_real	baseline	mmseqs2	9	48	0.188
ns_3826979_nc_421431_real	baseline	mummer4	15	48	0.312
ns_3826979_nc_421431_real	baseline	strobealign	6	48	0.125
ns_3826979_nc_421431_real	baseline	x_mapper	6	48	0.125
ns_3826979_nc_421431_real	baseline	blastn	316	2305	0.137
ns_3826979_nc_421431_real	baseline	bowtie1	2170	2305	0.941
ns_3826979_nc_421431_real	baseline	bowtie2	2124	2305	0.921
ns_3826979_nc_421431_real	baseline	indelfree_indexed	1353	2305	0.587
ns_3826979_nc_421431_real	baseline	mmseqs2	77	2305	0.033
ns_3826979_nc_421431_real	baseline	mummer4	95	2305	0.041
ns_3826979_nc_421431_real	baseline	strobealign	56	2305	0.024
ns_3826979_nc_421431_real	baseline	x_mapper	133	2305	0.058
ns_3826979_nc_421431_real	baseline	blastn	3086	53919	0.057
ns_3826979_nc_421431_real	baseline	bowtie1	52171	53919	0.968
ns_3826979_nc_421431_real	baseline	bowtie2	5282	53919	0.098
ns_3826979_nc_421431_real	baseline	indelfree_indexed	32407	53919	0.601
ns_3826979_nc_421431_real	baseline	mmseqs2	192	53919	0.004
ns_3826979_nc_421431_real	baseline	mummer4	186	53919	0.003
ns_3826979_nc_421431_real	baseline	strobealign	100	53919	0.002
ns_3826979_nc_421431_real	baseline	x_mapper	830	53919	0.015
ns_3826979_nc_421431_real	baseline	blastn	11675	15859	0.736

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
ns_3826979_nc_421431_real	baseline	bowtie2	273	15859	0.017
ns_3826979_nc_421431_real	baseline	indelfree_indexed	4	15859	0.000
ns_3826979_nc_421431_real	baseline	mmseqs2	183	15859	0.012
ns_3826979_nc_421431_real	baseline	mummer4	41	15859	0.003
ns_3826979_nc_421431_real	baseline	strobealign	13	15859	0.001
ns_3826979_nc_421431_real	baseline	x_mapper	3701	15859	0.233
ns_3826979_nc_421431_real	baseline	blastn	28199	41182	0.685
ns_3826979_nc_421431_real	baseline	bowtie2	555	41182	0.013
ns_3826979_nc_421431_real	baseline	mmseqs2	161	41182	0.004
ns_3826979_nc_421431_real	baseline	mummer4	27	41182	0.001
ns_3826979_nc_421431_real	baseline	x_mapper	12280	41182	0.298
ns_500000_nc_100000	0 edit	blastn	374003	374003	1.000
ns_500000_nc_100000	0 edit	bowtie1	374003	374003	1.000
ns_500000_nc_100000	0 edit	bowtie2	374003	374003	1.000
ns_500000_nc_100000	0 edit	indelfree_indexed	374003	374003	1.000
ns_500000_nc_100000	0 edit	minimap2	345	374003	0.001
ns_500000_nc_100000	0 edit	mmseqs2	372696	374003	0.997
ns_500000_nc_100000	0 edit	mummer4	374003	374003	1.000
ns_500000_nc_100000	0 edit	strobealign	349066	374003	0.933
ns_500000_nc_100000	1 edit	blastn	304580	402073	0.758
ns_500000_nc_100000	1 edit	bowtie1	375136	402073	0.933
ns_500000_nc_100000	1 edit	bowtie2	375282	402073	0.933
ns_500000_nc_100000	1 edit	indelfree_indexed	375136	402073	0.933
ns_500000_nc_100000	1 edit	mmseqs2	334553	402073	0.832
ns_500000_nc_100000	1 edit	mummer4	231562	402073	0.576
ns_500000_nc_100000	1 edit	strobealign	135486	402073	0.337
ns_500000_nc_100000	2 edit	blastn	259631	461711	0.562
ns_500000_nc_100000	2 edit	bowtie1	375951	461711	0.814
ns_500000_nc_100000	2 edit	bowtie2	377021	461711	0.817
ns_500000_nc_100000	2 edit	indelfree_indexed	375951	461711	0.814
ns_500000_nc_100000	2 edit	mmseqs2	241017	461711	0.522
ns_500000_nc_100000	2 edit	mummer4	112392	461711	0.243
ns_500000_nc_100000	2 edit	strobealign	38808	461711	0.084
ns_500000_nc_100000	3 edit	blastn	236102	534772	0.442
ns_500000_nc_100000	3 edit	bowtie1	388464	534772	0.726
ns_500000_nc_100000	3 edit	bowtie2	344571	534772	0.644
ns_500000_nc_100000	3 edit	indelfree_indexed	388464	534772	0.726
ns_500000_nc_100000	3 edit	mmseqs2	119599	534772	0.224
ns_500000_nc_100000	3 edit	mummer4	45357	534772	0.085
ns_500000_nc_100000	3 edit	strobealign	7391	534772	0.014
ns_500000_nc_100000	4 edit	blastn	123239	145864	0.845
ns_500000_nc_100000	4 edit	bowtie2	2	145864	0.000
ns_500000_nc_100000	4 edit	mmseqs2	32346	145864	0.222

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
ns_500000_nc_100000	4 edit	mummer4	22	145864	0.000
ns_500000_nc_100000	4 edit	strobealign	2	145864	0.000
ns_500000_nc_100000	5 edit	blastn	140734	145187	0.969
ns_500000_nc_100000	5 edit	mmseqs2	7907	145187	0.054
ns_500000_nc_100000	5 edit	mummer4	2	145187	0.000
ns_500000_nc_100000	0 hamming	blastn	374003	374003	1.000
ns_500000_nc_100000	0 hamming	bowtie1	374003	374003	1.000
ns_500000_nc_100000	0 hamming	bowtie2	374003	374003	1.000
ns_500000_nc_100000	0 hamming	indelfree_indexed	374003	374003	1.000
ns_500000_nc_100000	0 hamming	minimap2	345	374003	0.001
ns_500000_nc_100000	0 hamming	mmseqs2	372696	374003	0.997
ns_500000_nc_100000	0 hamming	mummer4	374003	374003	1.000
ns_500000_nc_100000	0 hamming	strobealign	349066	374003	0.933
ns_500000_nc_100000	1 hamming	blastn	304576	398710	0.764
ns_500000_nc_100000	1 hamming	bowtie1	375136	398710	0.941
ns_500000_nc_100000	1 hamming	bowtie2	375136	398710	0.941
ns_500000_nc_100000	1 hamming	indelfree_indexed	375136	398710	0.941
ns_500000_nc_100000	1 hamming	mmseqs2	331446	398710	0.831
ns_500000_nc_100000	1 hamming	mummer4	231354	398710	0.580
ns_500000_nc_100000	1 hamming	strobealign	135485	398710	0.340
ns_500000_nc_100000	2 hamming	blastn	258690	452616	0.572
ns_500000_nc_100000	2 hamming	bowtie1	375466	452616	0.830
ns_500000_nc_100000	2 hamming	bowtie2	375070	452616	0.829
ns_500000_nc_100000	2 hamming	indelfree_indexed	375466	452616	0.830
ns_500000_nc_100000	2 hamming	mmseqs2	234650	452616	0.518
ns_500000_nc_100000	2 hamming	mummer4	111399	452616	0.246
ns_500000_nc_100000	2 hamming	strobealign	38789	452616	0.086
ns_500000_nc_100000	3 hamming	blastn	232662	530594	0.438
ns_500000_nc_100000	3 hamming	bowtie1	388949	530594	0.733
ns_500000_nc_100000	3 hamming	bowtie2	344259	530594	0.649
ns_500000_nc_100000	3 hamming	indelfree_indexed	388949	530594	0.733
ns_500000_nc_100000	3 hamming	mmseqs2	119471	530594	0.225
ns_500000_nc_100000	3 hamming	mummer4	44531	530594	0.084
ns_500000_nc_100000	3 hamming	strobealign	7409	530594	0.014
ns_500000_nc_100000	4 hamming	blastn	114382	140204	0.816
ns_500000_nc_100000	4 hamming	bowtie2	653	140204	0.005
ns_500000_nc_100000	4 hamming	mmseqs2	34907	140204	0.249
ns_500000_nc_100000	4 hamming	mummer4	142	140204	0.001
ns_500000_nc_100000	4 hamming	strobealign	2	140204	0.000
ns_500000_nc_100000	5 hamming	blastn	110876	117208	0.946
ns_500000_nc_100000	5 hamming	bowtie2	386	117208	0.003
ns_500000_nc_100000	5 hamming	mmseqs2	9389	117208	0.080
ns_500000_nc_100000	5 hamming	mummer4	27	117208	0.000

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
ns_50000_nc_5000	0 edit	blastn	24794	24794	1.000
ns_50000_nc_5000	0 edit	bowtie1	24794	24794	1.000
ns_50000_nc_5000	0 edit	bowtie2	24794	24794	1.000
ns_50000_nc_5000	0 edit	indelfree_bruteforce	24794	24794	1.000
ns_50000_nc_5000	0 edit	indelfree_indexed	24794	24794	1.000
ns_50000_nc_5000	0 edit	minimap2	23	24794	0.001
ns_50000_nc_5000	0 edit	mmseqs2	24708	24794	0.997
ns_50000_nc_5000	0 edit	mummer4	24794	24794	1.000
ns_50000_nc_5000	0 edit	sassy	24794	24794	1.000
ns_50000_nc_5000	0 edit	strobealign	24469	24794	0.987
ns_50000_nc_5000	0 edit	x_mapper	24741	24794	0.998
ns_50000_nc_5000	1 edit	blastn	20401	27375	0.745
ns_50000_nc_5000	1 edit	bowtie1	25027	27375	0.914
ns_50000_nc_5000	1 edit	bowtie2	25036	27375	0.915
ns_50000_nc_5000	1 edit	indelfree_bruteforce	25027	27375	0.914
ns_50000_nc_5000	1 edit	indelfree_indexed	25027	27375	0.914
ns_50000_nc_5000	1 edit	mmseqs2	22412	27375	0.819
ns_50000_nc_5000	1 edit	mummer4	15564	27375	0.569
ns_50000_nc_5000	1 edit	sassy	25079	27375	0.916
ns_50000_nc_5000	1 edit	strobealign	18661	27375	0.682
ns_50000_nc_5000	1 edit	x_mapper	13084	27375	0.478
ns_50000_nc_5000	2 edit	blastn	17301	32010	0.540
ns_50000_nc_5000	2 edit	bowtie1	25081	32010	0.784
ns_50000_nc_5000	2 edit	bowtie2	25067	32010	0.783
ns_50000_nc_5000	2 edit	indelfree_bruteforce	25081	32010	0.784
ns_50000_nc_5000	2 edit	indelfree_indexed	25081	32010	0.784
ns_50000_nc_5000	2 edit	mmseqs2	16419	32010	0.513
ns_50000_nc_5000	2 edit	mummer4	7535	32010	0.235
ns_50000_nc_5000	2 edit	sassy	25240	32010	0.789
ns_50000_nc_5000	2 edit	strobealign	10733	32010	0.335
ns_50000_nc_5000	2 edit	x_mapper	6519	32010	0.204
ns_50000_nc_5000	3 edit	blastn	15362	37371	0.411
ns_50000_nc_5000	3 edit	bowtie1	24918	37371	0.667
ns_50000_nc_5000	3 edit	bowtie2	22707	37371	0.608
ns_50000_nc_5000	3 edit	indelfree_bruteforce	25084	37371	0.671
ns_50000_nc_5000	3 edit	indelfree_indexed	25084	37371	0.671
ns_50000_nc_5000	3 edit	mmseqs2	9326	37371	0.250
ns_50000_nc_5000	3 edit	mummer4	3003	37371	0.080
ns_50000_nc_5000	3 edit	sassy	25522	37371	0.683
ns_50000_nc_5000	3 edit	strobealign	4593	37371	0.123
ns_50000_nc_5000	3 edit	x_mapper	2401	37371	0.064
ns_50000_nc_5000	4 edit	blastn	14560	46390	0.314
ns_50000_nc_5000	4 edit	bowtie2	95	46390	0.002

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
ns_50000_nc_5000	4 edit	indelfree_bruteforce	25808	46390	0.556
ns_50000_nc_5000	4 edit	indelfree_indexed	25802	46390	0.556
ns_50000_nc_5000	4 edit	mmseqs2	4942	46390	0.107
ns_50000_nc_5000	4 edit	mummer4	214	46390	0.005
ns_50000_nc_5000	4 edit	sassy	31133	46390	0.671
ns_50000_nc_5000	4 edit	strobealign	1060	46390	0.023
ns_50000_nc_5000	4 edit	x_mapper	775	46390	0.017
ns_50000_nc_5000	5 edit	blastn	12016	154940	0.078
ns_50000_nc_5000	5 edit	indelfree_bruteforce	29297	154940	0.189
ns_50000_nc_5000	5 edit	indelfree_indexed	29109	154940	0.188
ns_50000_nc_5000	5 edit	mmseqs2	1891	154940	0.012
ns_50000_nc_5000	5 edit	mummer4	8	154940	0.000
ns_50000_nc_5000	5 edit	sassy	139124	154940	0.898
ns_50000_nc_5000	5 edit	strobealign	90	154940	0.001
ns_50000_nc_5000	5 edit	x_mapper	273	154940	0.002
ns_50000_nc_5000	0 hamming	blastn	24794	24794	1.000
ns_50000_nc_5000	0 hamming	bowtie1	24794	24794	1.000
ns_50000_nc_5000	0 hamming	bowtie2	24794	24794	1.000
ns_50000_nc_5000	0 hamming	indelfree_bruteforce	24794	24794	1.000
ns_50000_nc_5000	0 hamming	indelfree_indexed	24794	24794	1.000
ns_50000_nc_5000	0 hamming	minimap2	23	24794	0.001
ns_50000_nc_5000	0 hamming	mmseqs2	24708	24794	0.997
ns_50000_nc_5000	0 hamming	mummer4	24794	24794	1.000
ns_50000_nc_5000	0 hamming	sassy	24794	24794	1.000
ns_50000_nc_5000	0 hamming	strobealign	24469	24794	0.987
ns_50000_nc_5000	0 hamming	x_mapper	24741	24794	0.998
ns_50000_nc_5000	1 hamming	blastn	20401	26580	0.768
ns_50000_nc_5000	1 hamming	bowtie1	25027	26580	0.942
ns_50000_nc_5000	1 hamming	bowtie2	25027	26580	0.942
ns_50000_nc_5000	1 hamming	indelfree_bruteforce	25027	26580	0.942
ns_50000_nc_5000	1 hamming	indelfree_indexed	25027	26580	0.942
ns_50000_nc_5000	1 hamming	mmseqs2	22216	26580	0.836
ns_50000_nc_5000	1 hamming	mummer4	15543	26580	0.585
ns_50000_nc_5000	1 hamming	sassy	24467	26580	0.921
ns_50000_nc_5000	1 hamming	strobealign	18661	26580	0.702
ns_50000_nc_5000	1 hamming	x_mapper	13079	26580	0.492
ns_50000_nc_5000	2 hamming	blastn	17226	30709	0.561
ns_50000_nc_5000	2 hamming	bowtie1	25050	30709	0.816
ns_50000_nc_5000	2 hamming	bowtie2	25008	30709	0.814
ns_50000_nc_5000	2 hamming	indelfree_bruteforce	25050	30709	0.816
ns_50000_nc_5000	2 hamming	indelfree_indexed	25050	30709	0.816
ns_50000_nc_5000	2 hamming	mmseqs2	15894	30709	0.518
ns_50000_nc_5000	2 hamming	mummer4	7462	30709	0.243

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
ns_50000_nc_5000	2 hamming	sassy	24512	30709	0.798
ns_50000_nc_5000	2 hamming	strobealign	10730	30709	0.349
ns_50000_nc_5000	2 hamming	x_mapper	6505	30709	0.212
ns_50000_nc_5000	3 hamming	blastn	15094	35677	0.423
ns_50000_nc_5000	3 hamming	bowtie1	24949	35677	0.699
ns_50000_nc_5000	3 hamming	bowtie2	22689	35677	0.636
ns_50000_nc_5000	3 hamming	indelfree_bruteforce	24949	35677	0.699
ns_50000_nc_5000	3 hamming	indelfree_indexed	24949	35677	0.699
ns_50000_nc_5000	3 hamming	mmseqs2	9085	35677	0.255
ns_50000_nc_5000	3 hamming	mummer4	2933	35677	0.082
ns_50000_nc_5000	3 hamming	sassy	24379	35677	0.683
ns_50000_nc_5000	3 hamming	strobealign	4591	35677	0.129
ns_50000_nc_5000	3 hamming	x_mapper	2395	35677	0.067
ns_50000_nc_5000	4 hamming	blastn	13938	39393	0.354
ns_50000_nc_5000	4 hamming	bowtie2	143	39393	0.004
ns_50000_nc_5000	4 hamming	indelfree_bruteforce	25554	39393	0.649
ns_50000_nc_5000	4 hamming	indelfree_indexed	25549	39393	0.649
ns_50000_nc_5000	4 hamming	mmseqs2	4896	39393	0.124
ns_50000_nc_5000	4 hamming	mummer4	227	39393	0.006
ns_50000_nc_5000	4 hamming	sassy	24873	39393	0.631
ns_50000_nc_5000	4 hamming	strobealign	1062	39393	0.027
ns_50000_nc_5000	4 hamming	x_mapper	778	39393	0.020
ns_50000_nc_5000	5 hamming	blastn	10858	43594	0.249
ns_50000_nc_5000	5 hamming	bowtie2	34	43594	0.001
ns_50000_nc_5000	5 hamming	indelfree_bruteforce	29717	43594	0.682
ns_50000_nc_5000	5 hamming	indelfree_indexed	29528	43594	0.677
ns_50000_nc_5000	5 hamming	mmseqs2	2242	43594	0.051
ns_50000_nc_5000	5 hamming	mummer4	9	43594	0.000
ns_50000_nc_5000	5 hamming	sassy	28530	43594	0.654
ns_50000_nc_5000	5 hamming	strobealign	93	43594	0.002
ns_50000_nc_5000	5 hamming	x_mapper	288	43594	0.007
ns_500_nc_5000_HIGH_INSERTION_RATE	blastn	blastn	104817	104817	1.000
ns_500_nc_5000_HIGH_INSERTION_RATE	bowtie1	bowtie1	104817	104817	1.000
ns_500_nc_5000_HIGH_INSERTION_RATE	bowtie2	bowtie2	104817	104817	1.000
ns_500_nc_5000_HIGH_INSERTION_RATE	indelfree_bruteforce	indelfree_bruteforce	104817	104817	1.000
ns_500_nc_5000_HIGH_INSERTION_RATE	indelfree_indexed	indelfree_indexed	104817	104817	1.000
ns_500_nc_5000_HIGH_INSERTION_RATE	mmseqs2	mmseqs2	102984	104817	0.983
ns_500_nc_5000_HIGH_INSERTION_RATE	mummer4	mummer4	104817	104817	1.000
ns_500_nc_5000_HIGH_INSERTION_RATE	sassy	sassy	104817	104817	1.000
ns_500_nc_5000_HIGH_INSERTION_RATE	strobealign	strobealign	67040	104817	0.640
ns_500_nc_5000_HIGH_INSERTION_RATE	x_mapper	x_mapper	17184	104817	0.164
ns_500_nc_5000_HIGH_INSERTION_RATE	blastn	blastn	85929	114070	0.753
ns_500_nc_5000_HIGH_INSERTION_RATE	bowtie1	bowtie1	104853	114070	0.919

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
ns_500_nc_5000_HIGH_INSERTION_RATE	bowtie2	104853	114070	0.919	
ns_500_nc_5000_HIGH_INSERTION_RATE	ndelfree_bruteforce	104853	114070	0.919	
ns_500_nc_5000_HIGH_INSERTION_RATE	ndelfree_indexed	104853	114070	0.919	
ns_500_nc_5000_HIGH_INSERTION_RATE	nmseqs2	78927	114070	0.692	
ns_500_nc_5000_HIGH_INSERTION_RATE	nummer4	67178	114070	0.589	
ns_500_nc_5000_HIGH_INSERTION_RATE	easy	105363	114070	0.924	
ns_500_nc_5000_HIGH_INSERTION_RATE	etrobealign	10114	114070	0.089	
ns_500_nc_5000_HIGH_INSERTION_RATE	mapper	4538	114070	0.040	
ns_500_nc_5000_HIGH_INSERTION_RATE	elastn	70575	134564	0.524	
ns_500_nc_5000_HIGH_INSERTION_RATE	bowtie1	104879	134564	0.779	
ns_500_nc_5000_HIGH_INSERTION_RATE	bowtie2	105069	134564	0.781	
ns_500_nc_5000_HIGH_INSERTION_RATE	ndelfree_bruteforce	104879	134564	0.779	
ns_500_nc_5000_HIGH_INSERTION_RATE	ndelfree_indexed	104879	134564	0.779	
ns_500_nc_5000_HIGH_INSERTION_RATE	nmseqs2	29741	134564	0.221	
ns_500_nc_5000_HIGH_INSERTION_RATE	nummer4	31106	134564	0.231	
ns_500_nc_5000_HIGH_INSERTION_RATE	easy	105051	134564	0.781	
ns_500_nc_5000_HIGH_INSERTION_RATE	etrobealign	1600	134564	0.012	
ns_500_nc_5000_HIGH_INSERTION_RATE	mapper	1224	134564	0.009	
ns_500_nc_5000_HIGH_INSERTION_RATE	elastn	65189	151548	0.430	
ns_500_nc_5000_HIGH_INSERTION_RATE	bowtie1	104793	151548	0.691	
ns_500_nc_5000_HIGH_INSERTION_RATE	bowtie2	95824	151548	0.632	
ns_500_nc_5000_HIGH_INSERTION_RATE	ndelfree_bruteforce	105741	151548	0.698	
ns_500_nc_5000_HIGH_INSERTION_RATE	ndelfree_indexed	105741	151548	0.698	
ns_500_nc_5000_HIGH_INSERTION_RATE	nmseqs2	11718	151548	0.077	
ns_500_nc_5000_HIGH_INSERTION_RATE	nummer4	16857	151548	0.111	
ns_500_nc_5000_HIGH_INSERTION_RATE	easy	106486	151548	0.703	
ns_500_nc_5000_HIGH_INSERTION_RATE	etrobealign	151	151548	0.001	
ns_500_nc_5000_HIGH_INSERTION_RATE	elastn	61345	165498	0.371	
ns_500_nc_5000_HIGH_INSERTION_RATE	bowtie2	393	165498	0.002	
ns_500_nc_5000_HIGH_INSERTION_RATE	ndelfree_bruteforce	105081	165498	0.635	
ns_500_nc_5000_HIGH_INSERTION_RATE	ndelfree_indexed	105080	165498	0.635	
ns_500_nc_5000_HIGH_INSERTION_RATE	nmseqs2	5090	165498	0.031	
ns_500_nc_5000_HIGH_INSERTION_RATE	nummer4	7445	165498	0.045	
ns_500_nc_5000_HIGH_INSERTION_RATE	easy	107772	165498	0.651	
ns_500_nc_5000_HIGH_INSERTION_RATE	elastn	51390	167179	0.307	
ns_500_nc_5000_HIGH_INSERTION_RATE	ndelfree_bruteforce	104223	167179	0.623	
ns_500_nc_5000_HIGH_INSERTION_RATE	ndelfree_indexed	103972	167179	0.622	
ns_500_nc_5000_HIGH_INSERTION_RATE	nmseqs2	1429	167179	0.009	
ns_500_nc_5000_HIGH_INSERTION_RATE	nummer4	1780	167179	0.011	
ns_500_nc_5000_HIGH_INSERTION_RATE	easy	106805	167179	0.639	
ns_500_nc_5000_HIGH_INSERTION_RATE	elastn	104817	104817	1.000	
ns_500_nc_5000_HIGH_INSERTION_RATE	bowtie1	104817	104817	1.000	
ns_500_nc_5000_HIGH_INSERTION_RATE	bowtie2	104817	104817	1.000	

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
ns_500_nc_5000_HIGH_INSERTION_RATE	Endelfree_bruteforce	104817	104817	1.000	
ns_500_nc_5000_HIGH_INSERTION_RATE	Endelfree_indexed	104817	104817	1.000	
ns_500_nc_5000_HIGH_INSERTION_RATE	Emseqs2	102984	104817	0.983	
ns_500_nc_5000_HIGH_INSERTION_RATE	Enummer4	104817	104817	1.000	
ns_500_nc_5000_HIGH_INSERTION_RATE	Eassy	104817	104817	1.000	
ns_500_nc_5000_HIGH_INSERTION_RATE	Erobealign	67040	104817	0.640	
ns_500_nc_5000_HIGH_INSERTION_RATE	E_mapper	17184	104817	0.164	
ns_500_nc_5000_HIGH_INSERTION_RATE	Elastn	85929	110856	0.775	
ns_500_nc_5000_HIGH_INSERTION_RATE	Flowtie1	104853	110856	0.946	
ns_500_nc_5000_HIGH_INSERTION_RATE	Flowtie2	104853	110856	0.946	
ns_500_nc_5000_HIGH_INSERTION_RATE	Endelfree_bruteforce	104853	110856	0.946	
ns_500_nc_5000_HIGH_INSERTION_RATE	Endelfree_indexed	104853	110856	0.946	
ns_500_nc_5000_HIGH_INSERTION_RATE	Emseqs2	78212	110856	0.706	
ns_500_nc_5000_HIGH_INSERTION_RATE	Enummer4	67171	110856	0.606	
ns_500_nc_5000_HIGH_INSERTION_RATE	Eassy	102749	110856	0.927	
ns_500_nc_5000_HIGH_INSERTION_RATE	Erobealign	10114	110856	0.091	
ns_500_nc_5000_HIGH_INSERTION_RATE	E_mapper	4538	110856	0.041	
ns_500_nc_5000_HIGH_INSERTION_RATE	Elastn	70366	130166	0.541	
ns_500_nc_5000_HIGH_INSERTION_RATE	Flowtie1	104879	130166	0.806	
ns_500_nc_5000_HIGH_INSERTION_RATE	Flowtie2	104835	130166	0.805	
ns_500_nc_5000_HIGH_INSERTION_RATE	Endelfree_bruteforce	104879	130166	0.806	
ns_500_nc_5000_HIGH_INSERTION_RATE	Endelfree_indexed	104879	130166	0.806	
ns_500_nc_5000_HIGH_INSERTION_RATE	Emseqs2	28450	130166	0.219	
ns_500_nc_5000_HIGH_INSERTION_RATE	Enummer4	30880	130166	0.237	
ns_500_nc_5000_HIGH_INSERTION_RATE	Eassy	102215	130166	0.785	
ns_500_nc_5000_HIGH_INSERTION_RATE	Erobealign	1600	130166	0.012	
ns_500_nc_5000_HIGH_INSERTION_RATE	E_mapper	1224	130166	0.009	
ns_500_nc_5000_HIGH_INSERTION_RATE	Elastn	64023	146256	0.438	
ns_500_nc_5000_HIGH_INSERTION_RATE	Flowtie1	104793	146256	0.717	
ns_500_nc_5000_HIGH_INSERTION_RATE	Flowtie2	95520	146256	0.653	
ns_500_nc_5000_HIGH_INSERTION_RATE	Endelfree_bruteforce	104793	146256	0.717	
ns_500_nc_5000_HIGH_INSERTION_RATE	Endelfree_indexed	104793	146256	0.717	
ns_500_nc_5000_HIGH_INSERTION_RATE	Emseqs2	12135	146256	0.083	
ns_500_nc_5000_HIGH_INSERTION_RATE	Enummer4	16606	146256	0.114	
ns_500_nc_5000_HIGH_INSERTION_RATE	Eassy	102284	146256	0.699	
ns_500_nc_5000_HIGH_INSERTION_RATE	Erobealign	151	146256	0.001	
ns_500_nc_5000_HIGH_INSERTION_RATE	Elastn	58922	157761	0.373	
ns_500_nc_5000_HIGH_INSERTION_RATE	Flowtie2	574	157761	0.004	
ns_500_nc_5000_HIGH_INSERTION_RATE	Endelfree_bruteforce	105144	157761	0.666	
ns_500_nc_5000_HIGH_INSERTION_RATE	Endelfree_indexed	105144	157761	0.666	
ns_500_nc_5000_HIGH_INSERTION_RATE	Emseqs2	5826	157761	0.037	
ns_500_nc_5000_HIGH_INSERTION_RATE	Enummer4	6829	157761	0.043	
ns_500_nc_5000_HIGH_INSERTION_RATE	Eassy	101912	157761	0.646	

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
ns_500_nc_5000_HIGH_INSERTION_RATE	0 edit	blastn	46971	159669	0.294
ns_500_nc_5000_HIGH_INSERTION_RATE	0 edit	bowtie2	356	159669	0.002
ns_500_nc_5000_HIGH_INSERTION_RATE	0 edit	indelfree_bruteforce	105108	159669	0.658
ns_500_nc_5000_HIGH_INSERTION_RATE	0 edit	indelfree_indexed	104856	159669	0.657
ns_500_nc_5000_HIGH_INSERTION_RATE	0 edit	mmseqs2	1734	159669	0.011
ns_500_nc_5000_HIGH_INSERTION_RATE	0 edit	mummer4	1307	159669	0.008
ns_500_nc_5000_HIGH_INSERTION_RATE	0 edit	sassy	103158	159669	0.646
ns_75000_nc_5000	0 edit	blastn	37749	37749	1.000
ns_75000_nc_5000	0 edit	bowtie1	37749	37749	1.000
ns_75000_nc_5000	0 edit	bowtie2	37749	37749	1.000
ns_75000_nc_5000	0 edit	indelfree_bruteforce	37749	37749	1.000
ns_75000_nc_5000	0 edit	indelfree_indexed	37749	37749	1.000
ns_75000_nc_5000	0 edit	minimap2	46	37749	0.001
ns_75000_nc_5000	0 edit	mmseqs2	37625	37749	0.997
ns_75000_nc_5000	0 edit	mummer4	37749	37749	1.000
ns_75000_nc_5000	0 edit	sassy	37749	37749	1.000
ns_75000_nc_5000	0 edit	strobealign	37250	37749	0.987
ns_75000_nc_5000	0 edit	x_mapper	37677	37749	0.998
ns_75000_nc_5000	1 edit	blastn	30587	41184	0.743
ns_75000_nc_5000	1 edit	bowtie1	37597	41184	0.913
ns_75000_nc_5000	1 edit	bowtie2	37608	41184	0.913
ns_75000_nc_5000	1 edit	indelfree_bruteforce	37597	41184	0.913
ns_75000_nc_5000	1 edit	indelfree_indexed	37597	41184	0.913
ns_75000_nc_5000	1 edit	mmseqs2	33566	41184	0.815
ns_75000_nc_5000	1 edit	mummer4	23200	41184	0.563
ns_75000_nc_5000	1 edit	sassy	37680	41184	0.915
ns_75000_nc_5000	1 edit	strobealign	28038	41184	0.681
ns_75000_nc_5000	1 edit	x_mapper	19999	41184	0.486
ns_75000_nc_5000	2 edit	blastn	25622	47801	0.536
ns_75000_nc_5000	2 edit	bowtie1	37488	47801	0.784
ns_75000_nc_5000	2 edit	bowtie2	37467	47801	0.784
ns_75000_nc_5000	2 edit	indelfree_bruteforce	37488	47801	0.784
ns_75000_nc_5000	2 edit	indelfree_indexed	37488	47801	0.784
ns_75000_nc_5000	2 edit	mmseqs2	24672	47801	0.516
ns_75000_nc_5000	2 edit	mummer4	11190	47801	0.234
ns_75000_nc_5000	2 edit	sassy	37720	47801	0.789
ns_75000_nc_5000	2 edit	strobealign	16043	47801	0.336
ns_75000_nc_5000	2 edit	x_mapper	9797	47801	0.205
ns_75000_nc_5000	3 edit	blastn	23305	56388	0.413
ns_75000_nc_5000	3 edit	bowtie1	37288	56388	0.661
ns_75000_nc_5000	3 edit	bowtie2	33991	56388	0.603
ns_75000_nc_5000	3 edit	indelfree_bruteforce	37471	56388	0.665
ns_75000_nc_5000	3 edit	indelfree_indexed	37471	56388	0.665

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
ns_75000_nc_5000	3 edit	mmseqs2	14147	56388	0.251
ns_75000_nc_5000	3 edit	mummer4	4386	56388	0.078
ns_75000_nc_5000	3 edit	sassy	38213	56388	0.678
ns_75000_nc_5000	3 edit	strobealign	6684	56388	0.119
ns_75000_nc_5000	3 edit	x_mapper	3684	56388	0.065
ns_75000_nc_5000	4 edit	blastn	21684	69095	0.314
ns_75000_nc_5000	4 edit	bowtie2	153	69095	0.002
ns_75000_nc_5000	4 edit	indelfree_bruteforce	38340	69095	0.555
ns_75000_nc_5000	4 edit	indelfree_indexed	38326	69095	0.555
ns_75000_nc_5000	4 edit	mmseqs2	7359	69095	0.107
ns_75000_nc_5000	4 edit	mummer4	313	69095	0.005
ns_75000_nc_5000	4 edit	sassy	46198	69095	0.669
ns_75000_nc_5000	4 edit	strobealign	1604	69095	0.023
ns_75000_nc_5000	4 edit	x_mapper	1142	69095	0.017
ns_75000_nc_5000	5 edit	blastn	18148	230913	0.079
ns_75000_nc_5000	5 edit	indelfree_bruteforce	44419	230913	0.192
ns_75000_nc_5000	5 edit	indelfree_indexed	44109	230913	0.191
ns_75000_nc_5000	5 edit	mmseqs2	2915	230913	0.013
ns_75000_nc_5000	5 edit	mummer4	7	230913	0.000
ns_75000_nc_5000	5 edit	sassy	206911	230913	0.896
ns_75000_nc_5000	5 edit	strobealign	130	230913	0.001
ns_75000_nc_5000	5 edit	x_mapper	391	230913	0.002
ns_75000_nc_5000	0 hamming	blastn	37749	37749	1.000
ns_75000_nc_5000	0 hamming	bowtie1	37749	37749	1.000
ns_75000_nc_5000	0 hamming	bowtie2	37749	37749	1.000
ns_75000_nc_5000	0 hamming	indelfree_bruteforce	37749	37749	1.000
ns_75000_nc_5000	0 hamming	indelfree_indexed	37749	37749	1.000
ns_75000_nc_5000	0 hamming	minimap2	46	37749	0.001
ns_75000_nc_5000	0 hamming	mmseqs2	37625	37749	0.997
ns_75000_nc_5000	0 hamming	mummer4	37749	37749	1.000
ns_75000_nc_5000	0 hamming	sassy	37749	37749	1.000
ns_75000_nc_5000	0 hamming	strobealign	37250	37749	0.987
ns_75000_nc_5000	0 hamming	x_mapper	37677	37749	0.998
ns_75000_nc_5000	1 hamming	blastn	30587	40038	0.764
ns_75000_nc_5000	1 hamming	bowtie1	37597	40038	0.939
ns_75000_nc_5000	1 hamming	bowtie2	37597	40038	0.939
ns_75000_nc_5000	1 hamming	indelfree_bruteforce	37597	40038	0.939
ns_75000_nc_5000	1 hamming	indelfree_indexed	37597	40038	0.939
ns_75000_nc_5000	1 hamming	mmseqs2	33244	40038	0.830
ns_75000_nc_5000	1 hamming	mummer4	23179	40038	0.579
ns_75000_nc_5000	1 hamming	sassy	36835	40038	0.920
ns_75000_nc_5000	1 hamming	strobealign	28038	40038	0.700
ns_75000_nc_5000	1 hamming	x_mapper	19998	40038	0.499

Fraction/Simulation Distance	Metric	Tool	n_found	n_total	Recall
ns_75000_nc_5000	2 hamming	blastn	25537	45848	0.557
ns_75000_nc_5000	2 hamming	bowtie1	37440	45848	0.817
ns_75000_nc_5000	2 hamming	bowtie2	37369	45848	0.815
ns_75000_nc_5000	2 hamming	indelfree_bruteforce	37440	45848	0.817
ns_75000_nc_5000	2 hamming	indelfree_indexed	37440	45848	0.817
ns_75000_nc_5000	2 hamming	mmseqs2	23882	45848	0.521
ns_75000_nc_5000	2 hamming	mummer4	11093	45848	0.242
ns_75000_nc_5000	2 hamming	sassy	36597	45848	0.798
ns_75000_nc_5000	2 hamming	strobealign	16034	45848	0.350
ns_75000_nc_5000	2 hamming	x_mapper	9780	45848	0.213
ns_75000_nc_5000	3 hamming	blastn	22922	53789	0.426
ns_75000_nc_5000	3 hamming	bowtie1	37336	53789	0.694
ns_75000_nc_5000	3 hamming	bowtie2	33946	53789	0.631
ns_75000_nc_5000	3 hamming	indelfree_bruteforce	37336	53789	0.694
ns_75000_nc_5000	3 hamming	indelfree_indexed	37336	53789	0.694
ns_75000_nc_5000	3 hamming	mmseqs2	13759	53789	0.256
ns_75000_nc_5000	3 hamming	mummer4	4302	53789	0.080
ns_75000_nc_5000	3 hamming	sassy	36445	53789	0.678
ns_75000_nc_5000	3 hamming	strobealign	6684	53789	0.124
ns_75000_nc_5000	3 hamming	x_mapper	3675	53789	0.068
ns_75000_nc_5000	4 hamming	blastn	20783	58871	0.353
ns_75000_nc_5000	4 hamming	bowtie2	243	58871	0.004
ns_75000_nc_5000	4 hamming	indelfree_bruteforce	37904	58871	0.644
ns_75000_nc_5000	4 hamming	indelfree_indexed	37898	58871	0.644
ns_75000_nc_5000	4 hamming	mmseqs2	7362	58871	0.125
ns_75000_nc_5000	4 hamming	mummer4	330	58871	0.006
ns_75000_nc_5000	4 hamming	sassy	36959	58871	0.628
ns_75000_nc_5000	4 hamming	strobealign	1610	58871	0.027
ns_75000_nc_5000	4 hamming	x_mapper	1140	58871	0.019
ns_75000_nc_5000	5 hamming	blastn	16383	65820	0.249
ns_75000_nc_5000	5 hamming	bowtie2	53	65820	0.001
ns_75000_nc_5000	5 hamming	indelfree_bruteforce	45038	65820	0.684
ns_75000_nc_5000	5 hamming	indelfree_indexed	44720	65820	0.679
ns_75000_nc_5000	5 hamming	mmseqs2	3419	65820	0.052
ns_75000_nc_5000	5 hamming	mummer4	12	65820	0.000
ns_75000_nc_5000	5 hamming	sassy	43018	65820	0.654
ns_75000_nc_5000	5 hamming	strobealign	132	65820	0.002
ns_75000_nc_5000	5 hamming	x_mapper	413	65820	0.006

1.1.7 Supplementary Table S9: Computational Resource Usage

Detailed resource usage metrics for all completed tool runs on both real (IMG/VR4 subsampled) and simulated datasets. For each tool-dataset combination, we report total CPU time (hours), wall-clock time (hours), peak memory usage (GB), SLURM job ID, and allocated CPU cores. All tools were allocated 64 threads

and 512 GB RAM on identical hardware (dual AMD EPYC 7543, see Methods the Reproducibility section in the main text).

The fraction_X datasets represent subsampled fractions of IMG/VR4 HQ contigs, while ns_X_nc_Y datasets are synthetic benchmarks. The ns_3826979_nc_421431_real_baseline dataset is semi-synthetic, where real spacers are searched against simulated contigs matched to IMG/VR4 characteristics.

1.1.7.1 A. Real data (IMG/VR4 subsampled fractions) — completed jobs

Tool	Fraction	Job ID	CPU Time (h)	Wall Time (h)	Peak Mem (GB)	CPUs	Max Distance
blastn	0.0005	20697879	6.60	0.55	0.61	38	perc_id=84%
blastn	0.001	20697880	9.02	0.55	0.63	38	perc_id=84%
blastn	0.005	20697881	44.79	3.19	0.68	38	perc_id=84%
blastn	0.01	20697882	94.72	9.00	0.70	38	perc_id=84%
blastn	0.05	20697883	556.64	28.77	1.31	38	perc_id=84%
blastn	0.1	20697884	1135.98	146.80	2.82	38	perc_id=84%
bowtie1	0.0005	20668861	0.04	0.04	0.18	38	3
bowtie1	0.001	20668609	0.04	0.04	0.50	38	3
bowtie1	0.005	20668621	0.09	0.09	0.83	38	3
bowtie1	0.01	20668633	0.11	0.11	1.22	38	3
bowtie1	0.05	20668645	4.07	0.32	4.15	38	3
bowtie1	0.1	20668655	10.27	1.50	8.02	38	3
bowtie1	1.0	20668665	117.29	14.33	146.37	38	3
bowtie2	0.0005	20668433	0.06	0.06	1.05	38	N/A
bowtie2	0.001	20668610	0.06	0.06	0.78	38	N/A
bowtie2	0.005	20668622	0.16	0.16	1.10	38	N/A
bowtie2	0.01	20668634	2.88	0.39	1.84	38	N/A
bowtie2	0.05	20668646	13.63	1.46	6.28	38	N/A
bowtie2	0.1	20668656	30.74	3.40	11.62	38	N/A
bowtie2	1.0	20668666	994.87	115.92	151.48	38	N/A
indelfree_bruc	0.0005	20668599	197.77	26.75	22.04	38	N/A
indelfree_bruc	0.001	20668611	371.13	41.72	27.82	38	N/A
indelfree_index	0.0005	20668600	2.72	0.47	20.31	38	N/A
indelfree_index	0.001	20668612	4.06	0.69	27.88	38	N/A
indelfree_index	0.005	20668624	20.73	2.54	28.42	38	N/A
indelfree_index	0.01	20668636	44.38	6.04	31.41	38	N/A
indelfree_index	0.05	20668647	185.84	18.43	35.83	38	N/A
indelfree_index	0.1	20668657	418.20	51.18	40.76	38	N/A
minimap2	0.0005	20668602	0.03	0.03	0.76	38	N/A
minimap2	0.001	20668614	0.04	0.04	0.76	38	N/A
minimap2	0.005	20668626	0.15	0.15	1.12	38	N/A
minimap2	0.01	20668638	2.61	0.63	1.65	38	N/A
minimap2	0.05	20668649	15.64	0.92	5.15	38	N/A
minimap2	0.1	20668659	34.19	7.72	9.85	38	N/A
minimap2	1.0	20668669	517.35	48.13	90.69	38	N/A

Tool	Fraction	Job ID	CPU Time (h)	Wall Time (h)	Peak Mem (GB)	CPUs	Max Distance
mmseqs2	0.0005	20668603	0.07	0.07	9.99	38	min-seq- id=85%
mmseqs2	0.001	20668615	0.07	0.07	9.98	38	min-seq- id=85%
mmseqs2	0.005	20668627	0.07	0.07	10.33	38	min-seq- id=85%
mmseqs2	0.01	20668639	0.07	0.07	10.75	38	min-seq- id=85%
mmseqs2	0.05	20668650	0.07	0.07	15.30	38	min-seq- id=85%
mmseqs2	0.1	20668660	0.13	0.13	22.84	38	min-seq- id=85%
mmseqs2	1.0	20668670	5.37	0.73	141.76	38	min-seq- id=85%
mummer4	0.0005	20668604	0.01	0.01	0.39	38	N/A
mummer4	0.001	20668616	0.01	0.01	0.12	38	N/A
mummer4	0.005	20668628	0.03	0.03	0.26	38	N/A
mummer4	0.01	20668640	0.08	0.08	0.28	38	N/A
mummer4	0.05	20668651	0.26	0.26	0.43	38	N/A
mummer4	0.1	20668661	4.64	0.86	2.31	38	N/A
mummer4	1.0	20668671	47.88	9.33	9.02	38	N/A
sassy	0.0005	20668605	26.24	2.75	2.95	38	N/A
sassy	0.001	20668617	37.25	4.66	3.14	38	N/A
sassy	0.005	20668629	212.28	25.47	10.43	38	N/A
sassy	0.01	20668641	439.60	65.63	27.94	38	N/A
strobealign	0.0005	20668606	0.01	0.01	0.63	38	N/A
strobealign	0.001	20668618	0.00	0.00	0.67	38	N/A
strobealign	0.005	20668630	0.01	0.01	2.02	38	N/A
strobealign	0.01	20668642	0.02	0.02	3.55	38	N/A
strobealign	0.05	20668652	0.03	0.03	11.72	38	N/A
strobealign	0.1	20668662	0.08	0.08	21.29	38	N/A
strobealign	1.0	20668672	0.32	0.32	116.67	38	N/A
x_mapper	0.0005	20668607	0.02	0.02	6.04	38	N/A
x_mapper	0.001	20668619	0.03	0.03	6.07	38	N/A
x_mapper	0.005	20668631	0.06	0.06	4.18	38	N/A
x_mapper	0.01	20668643	0.06	0.06	7.35	38	N/A
x_mapper	0.05	20668653	0.43	0.43	10.75	38	N/A
x_mapper	0.1	20668663	3.36	1.11	10.86	38	N/A
x_mapper	1.0	20695756	45.50	12.42	244.28	38	N/A

1.1.7.2 B. Simulated datasets — completed jobs

				N	N			CPU	Wall	Peak		Max
Dataset	Tool	Job ID	Search Space	Spacers	Contigs	Spacer Mbp	Contig Mbp	Time (h)	Time (h)	Mem (GB)	CPUs	Distance
ns_100000	hstc	20097887	10000000000000	100000	10000	3.20	802.83	12.99	1.11	1.14	38	perc_id=84%
ns_100000	wtic1	20097887	10000000000000	100000	10000	3.20	802.83	7.09	0.45	5.69	38	3
ns_100000	wtic2	20097891	10000000000000	100000	10000	3.20	802.83	3.83	0.48	6.52	38	N/A
ns_100000	hfree	20097891	10000000000000	100000	10000	3.20	802.83	653.95	51.71	105.35	38	5
ns_100000	hfree	20097894	10000000000000	100000	10000	3.20	802.83	6.24	0.26	39.30	38	5
ns_100000	inmap	20097895	10000000000000	100000	10000	3.20	802.83	0.07	0.07	5.33	38	N/A
ns_100000	inseqs	20097897	10000000000000	100000	10000	3.20	802.83	0.06	0.06	14.13	38	min-seq-id=85%
ns_100000	inmer	20097898	10000000000000	100000	10000	3.20	802.83	0.08	0.08	0.27	38	N/A
ns_100000	ssync	20097901	10000000000000	100000	10000	3.20	802.83	68.36	7.42	0.12	38	5
ns_100000	roheal	20097902	10000000000000	100000	10000	3.20	802.83	0.01	0.01	3.80	38	N/A
ns_100000	map	20097903	10000000000000	100000	10000	3.20	802.83	0.08	0.08	21.95	38	N/A
ns_100000	hstc	20097905	20000000000000	200000	20000	3.20	1604.14	25.74	1.14	2.13	38	perc_id=84%
ns_100000	wtic1	20097906	20000000000000	200000	20000	3.20	1604.14	6.92	0.70	8.13	38	3
ns_100000	wtic2	20097908	20000000000000	200000	20000	3.20	1604.14	7.41	0.76	12.21	38	N/A
ns_100000	hfree	20097909	20000000000000	200000	20000	3.20	1604.14	2948.61	137.93	32.38	38	5
ns_100000	hfree	20097910	20000000000000	200000	20000	3.20	1604.14	13.02	1.55	29.51	38	5
ns_100000	inmap	20097912	20000000000000	200000	20000	3.20	1604.14	0.05	0.05	9.98	38	N/A
ns_100000	inseqs	20097912	20000000000000	200000	20000	3.20	1604.14	0.04	0.04	20.16	38	min-seq-id=85%
ns_100000	inmer	20097913	20000000000000	200000	20000	3.20	1604.14	0.16	0.16	0.28	38	N/A
ns_100000	ssync	20097914	20000000000000	200000	20000	3.20	1604.14	141.30	5.74	0.15	38	5
ns_100000	roheal	20097915	20000000000000	200000	20000	3.20	1604.14	0.02	0.02	7.58	38	N/A
ns_100000	map	20097916	20000000000000	200000	20000	3.20	1604.14	2.16	0.42	29.34	38	N/A
ns_3826979	wtic1	20097307	16428075890697	421431	129.49	4291.11	109.49	10.43	182.05		38	3
ns_3826979	wtic2	20097308	16428075890697	421431	129.49	4291.11	618.22	49.78	164.30		38	N/A
ns_3826979	inmap	20097309	16428075890697	421431	129.49	4291.11	451.04	27.94	77.42		38	N/A
ns_3826979	inseqs	20097310	16428075890697	421431	129.49	4291.11	6.55	0.72	172.10		38	min-seq-id=85%
ns_3826979	inmer	20097311	16428075890697	421431	129.49	4291.11	72.12	12.51	20.54		38	N/A
ns_3826979	roheal	20097312	16428075890697	421431	129.49	4291.11	0.23	0.23	93.29		38	N/A
ns_3826979	map	20097313	16428075890697	421431	129.49	4291.11	32.10	2.87	229.82		38	N/A
ns_500000	hstc	20097915	50000000000000	500000	100000	16.00	2232.68	2578.12	208.46	33.08	38	perc_id=84%
ns_500000	wtic1	20097916	50000000000000	500000	100000	16.00	2232.68	131.26	24.04	207.01	38	3
ns_500000	wtic2	20097917	50000000000000	500000	100000	16.00	2232.68	210.24	26.82	208.15	38	N/A
ns_500000	hfree	20097920	50000000000000	500000	100000	16.00	2232.68	414.51	40.23	70.63	38	3
ns_500000	inmap	20097921	50000000000000	500000	100000	16.00	2232.68	70.34	3.22	89.59	38	N/A
ns_500000	inseqs	20097922	50000000000000	500000	100000	16.00	2232.68	8.30	1.23	217.16	38	min-seq-id=85%

Dataset	Tool	Job ID	Search Space	N Spacers	N Contigs	Spacer Mbp	Contig Mbp	CPU Time (h)	Wall Time (h)	Peak Mem (GB)	CPUs	Max Distance
ns_50000	humm	5200697923	5000000000000	100000	16.00	2232.68	14.59	3.26	1.89	38	N/A	
ns_50000	strocal	5200697924	5000000000000	100000	16.00	2232.68	0.23	0.23	141.83	38	N/A	
ns_50000	blastn	5200697925	5000000000000	5000	1.60	402.61	3.35	0.18	0.70	38	perc_id=84%	
ns_50000	bowtie	5200704672	5000000000000	5000	1.60	402.61	3.71	0.21	3.29	38	3	
ns_50000	bowtie	5200697928	5000000000000	5000	1.60	402.61	0.19	0.19	3.77	38	N/A	
ns_50000	ndbfr	5200697929	5000000000000	5000	1.60	402.61	251.32	10.48	85.46	38	5	
ns_50000	ndbfr	5200697930	5000000000000	5000	1.60	402.61	0.10	0.10	66.20	38	5	
ns_50000	minimap	5200697932	5000000000000	5000	1.60	402.61	0.01	0.01	2.64	38	N/A	
ns_50000	mmseqs	5200697932	5000000000000	5000	1.60	402.61	0.02	0.02	11.29	38	min-seq-id=85%	
ns_50000	humm	5200697933	5000000000000	5000	1.60	402.61	0.04	0.04	0.26	38	N/A	
ns_50000	ssync	5200697934	5000000000000	5000	1.60	402.61	18.18	1.06	0.06	38	5	
ns_50000	strocal	5200697935	5000000000000	5000	1.60	402.61	0.00	0.00	2.27	38	N/A	
ns_50000	map	5200697935	5000000000000	5000	1.60	402.61	0.04	0.04	15.75	38	N/A	
ns_500	blast	5002069786	5000000000000	5000	0.02	400.78	0.02	0.02	1.46	38	perc_id=84%	
ns_500	bowt	5002070467	5000000000000	5000	0.02	400.78	2.99	0.21	3.64	38	3	
ns_500	bowt	5002069787	5000000000000	5000	0.02	400.78	0.18	0.18	4.09	38	N/A	
ns_500	indel	5002069788	5000000000000	5000	0.02	400.78	0.18	0.18	27.47	38	5	
ns_500	indel	5002069789	5000000000000	5000	0.02	400.78	0.02	0.02	7.56	38	5	
ns_500	mini	5002069789	5000000000000	5000	0.02	400.78	0.01	0.01	2.69	38	N/A	
ns_500	mm	5002069789	5000000000000	5000	0.02	400.78	0.04	0.04	11.21	38	min-seq-id=85%	
ns_500	man	5002069789	5000000000000	5000	0.02	400.78	0.07	0.07	0.28	38	N/A	
ns_500	ssync	5002069789	5000000000000	5000	0.02	400.78	0.03	0.03	0.09	38	5	
ns_500	stro	5002069789	5000000000000	5000	0.02	400.78	0.00	0.00	1.85	38	N/A	
ns_500	mc	5002069789	5000000000000	5000	0.02	400.78	0.07	0.07	13.16	38	N/A	
ns_75000	blastn	5200697937	7500000000000	5000	2.40	401.07	4.88	0.30	0.69	38	perc_id=84%	
ns_75000	bowtie	5200704674	7500000000000	5000	2.40	401.07	3.01	0.20	3.67	38	3	
ns_75000	bowtie	5200697938	7500000000000	5000	2.40	401.07	0.18	0.18	3.72	38	N/A	
ns_75000	ndbfr	5200697940	7500000000000	5000	2.40	401.07	407.00	16.98	85.47	38	5	
ns_75000	ndbfr	5200697941	7500000000000	5000	2.40	401.07	2.54	0.14	85.13	38	5	
ns_75000	minimap	5200697942	7500000000000	5000	2.40	401.07	0.02	0.02	2.63	38	N/A	
ns_75000	mmseqs	5200697943	7500000000000	5000	2.40	401.07	0.02	0.02	11.44	38	min-seq-id=85%	
ns_75000	humm	5200697944	7500000000000	5000	2.40	401.07	0.04	0.04	0.26	38	N/A	
ns_75000	ssync	5200697943	7500000000000	5000	2.40	401.07	25.83	1.36	0.08	38	5	
ns_75000	strocal	5200697943	7500000000000	5000	2.40	401.07	0.01	0.01	2.31	38	N/A	

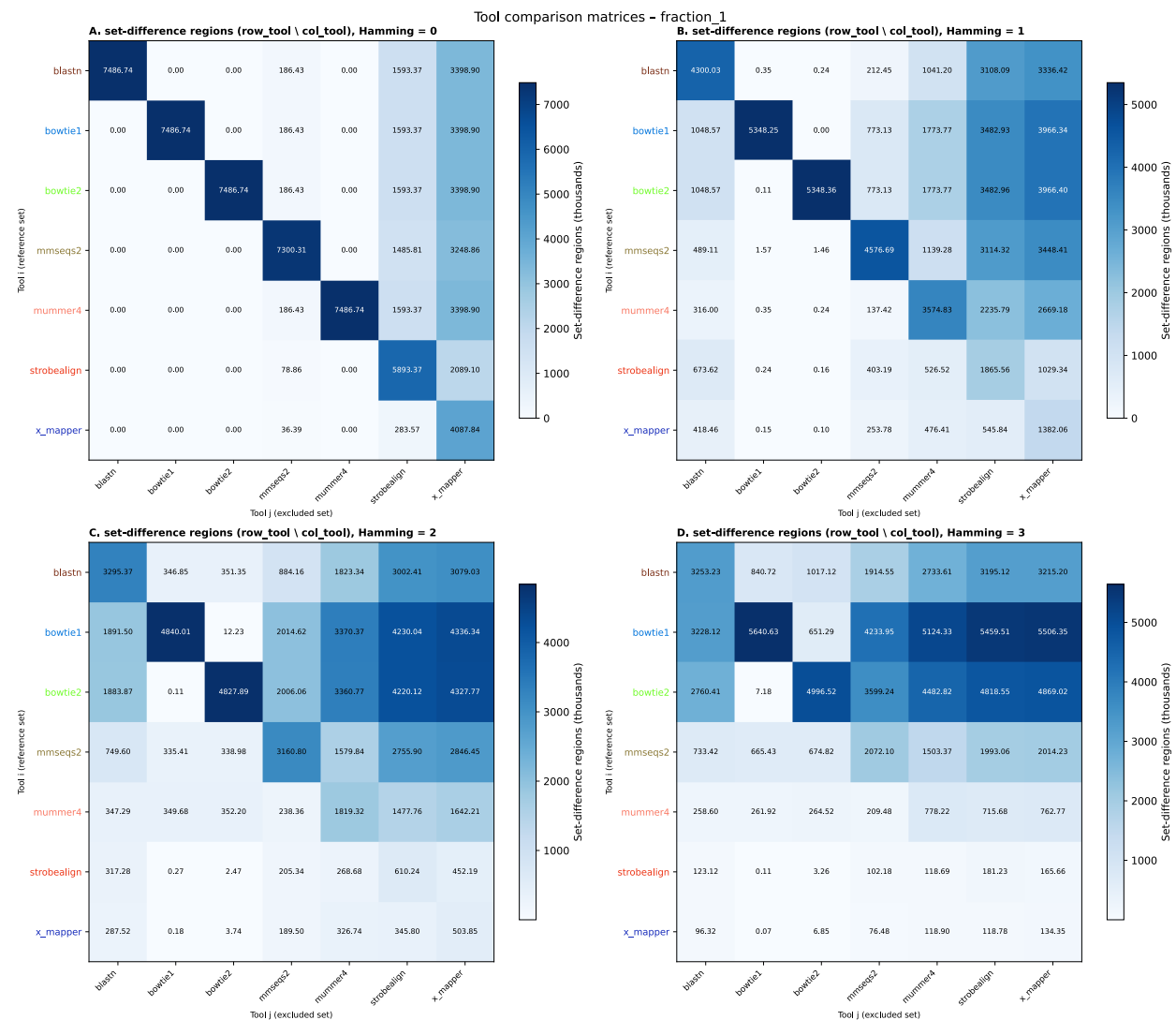
1.1.7.3 C. Timed-out / failed jobs

Tool	Fraction	Status	Job ID	CPUs	Time (h)
indelfree_bruteforce	0.005	TIMEOUT	20668623	38	216.01
indelfree_bruteforce	0.01	TIMEOUT	20668635	38	216.03
indelfree_indexed	1.0	TIMEOUT	20668667	38	216.00

1.2 Supplementary Figures

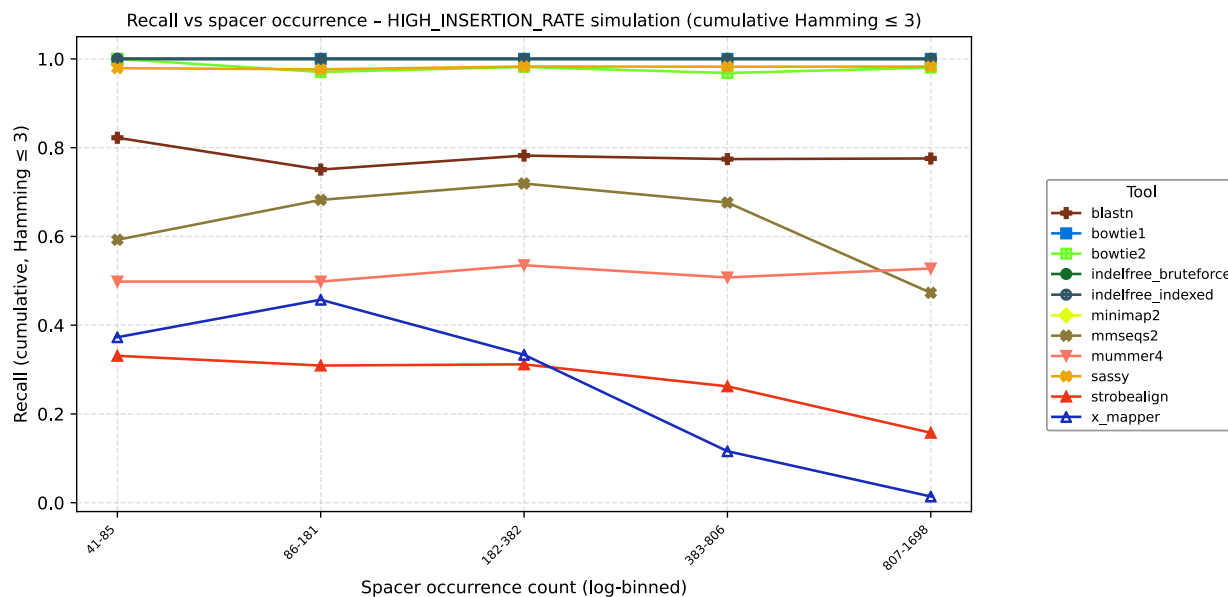
1.2.1 Supplementary Figure S1: IMG/VR4 tool-vs-tool comparison matrices

Pairwise tool-comparison matrices for the IMG/VR4 fraction_1 dataset at exact hamming distance values. Each sub-panel corresponds to a specific hamming distance (0, 1, 2, 3). Cell (i, j) shows the number of unique spacer-contig aligned regions detected by tool i (row) but not by tool j (column) at that exact distance. The diagonal cells (where $i = j$) show the total number of unique spacer-contig aligned regions detected by each tool at that distance.



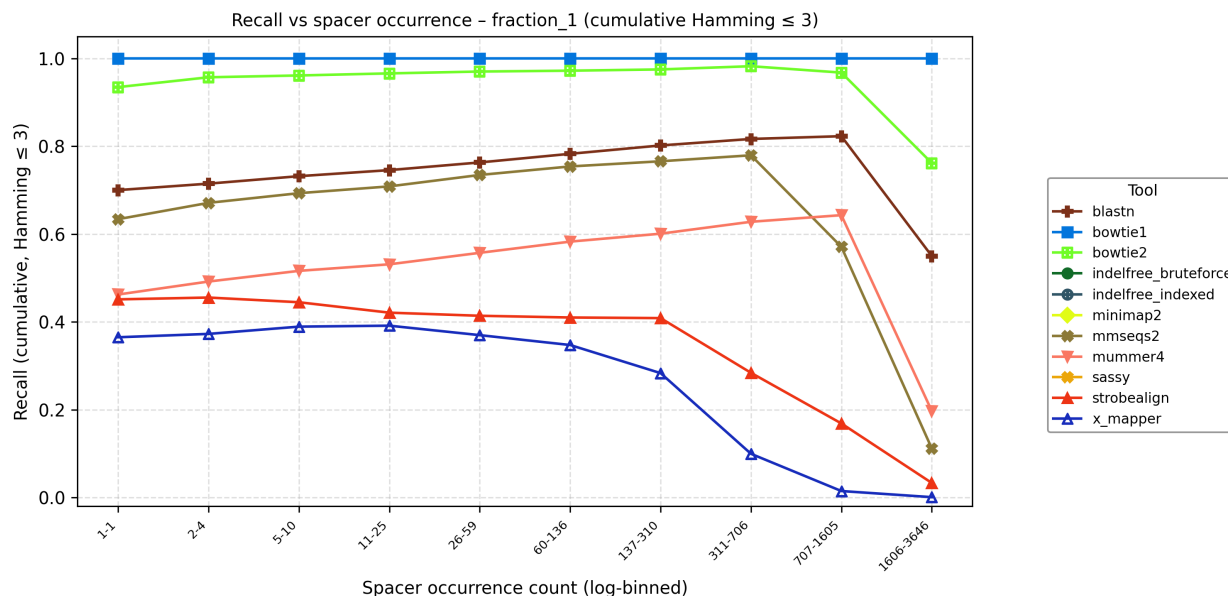
1.2.2 Supplementary Figure S2: Simulated dataset recall vs spacer occurrence frequency

High-occurrence-focused recall view (detection fraction) for the high-insertion simulated dataset (ns_500_nc_5000_HIGH_INSERTION_RATE) at hamming distance 3.



1.2.3 Supplementary Figure S3: IMG/VR4 fraction_1 recall vs occurrence frequency at hamming distance 3

High-occurrence-focused recall view for the real IMG/VR4 fraction_1 dataset at hamming distance 3.

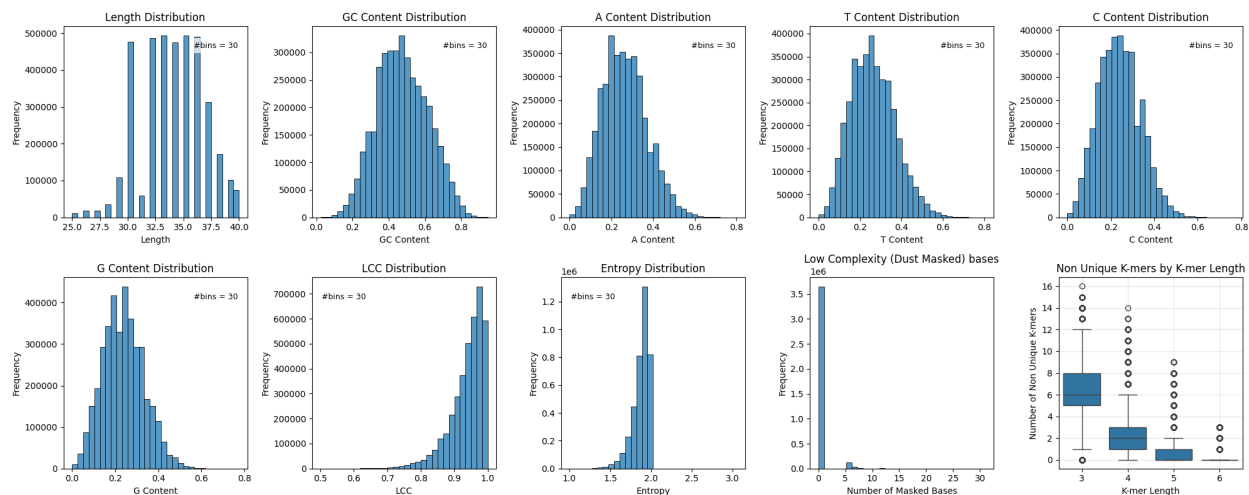


1.2.4 Supplementary Figure S4: Comparison of simulated vs real spacer characteristics

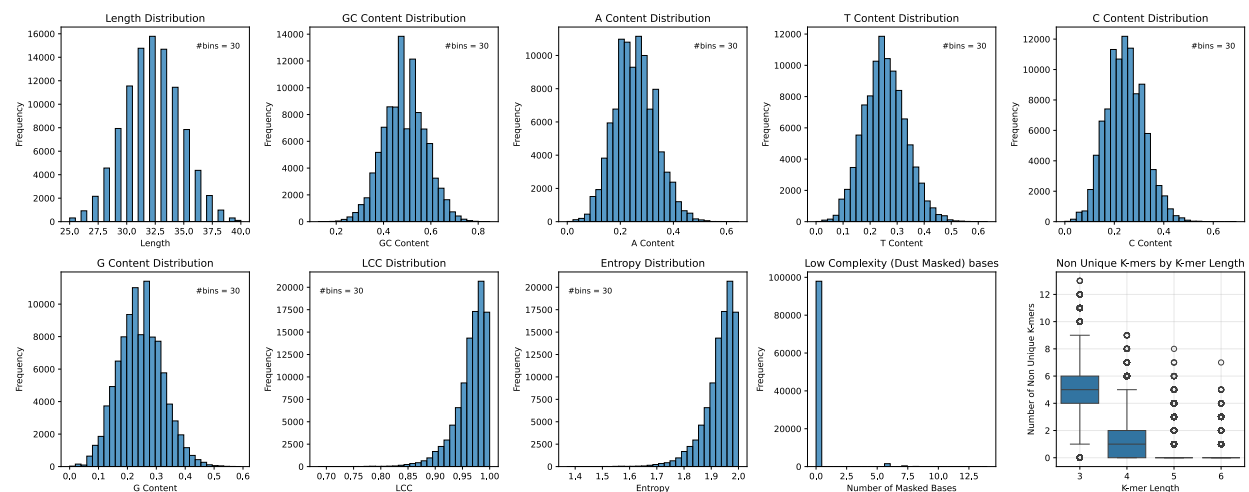
Distributions of spacer sequence features comparing simulated and real datasets, demonstrating that synthetic sequences approximate real data characteristics. Features analyzed include k-mer repeatability, Shannon entropy, base frequencies, GC content, and Local Composition Complexity (LCC, see

<https://doi.org/10.1038/npg.els.0005260> and <https://github.com/biopython/biopython/blob/e451db211bdd855a5d0f1f6bba1> for details).

A. Filtered iPHoP spacer set:



B. Fully simulated spacer set (dataset: ns_100000_nc_20000):



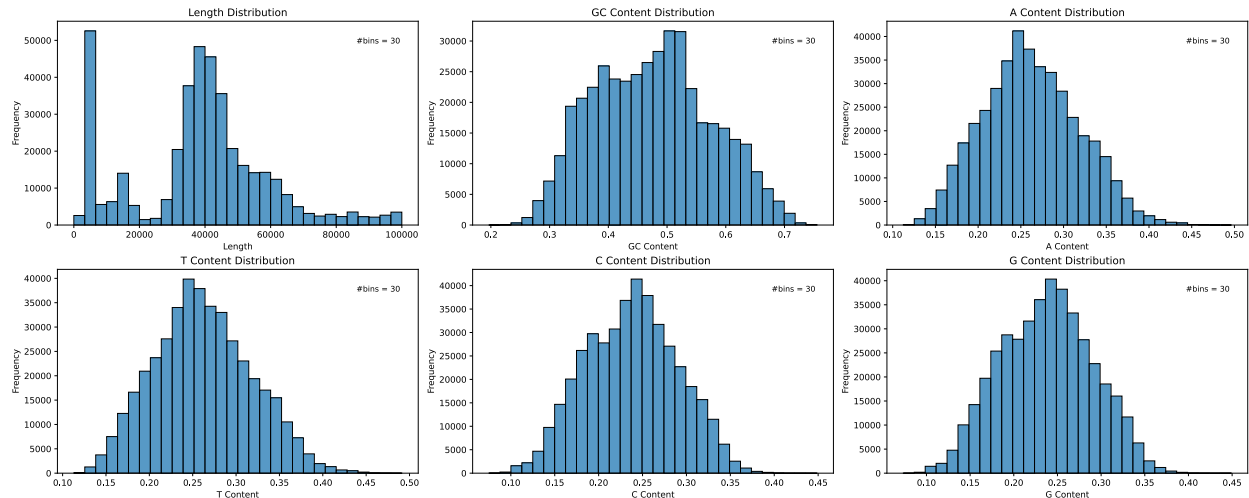
1.2.5 Supplementary Figure S5: Spacer characteristic distributions by dataset

Distributions of spacers characteristics for both real (IMG/VR4) and simulated datasets.

- (A) Spacer size (length in bp)
- (B) Mismatches observed in spacer alignments
- (C) Spacer occurrence rate.

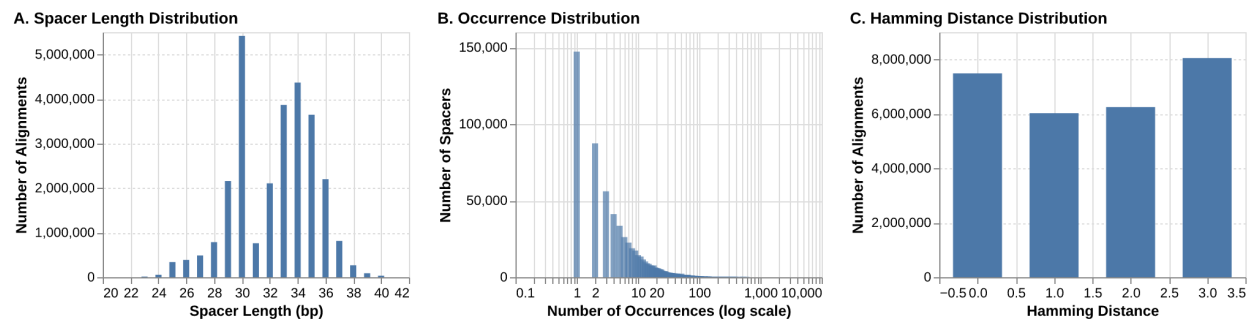
1.2.5.1 1. IMG/VR4 dataset (contig-side distributions)

Note: we removed outliers to enable plotting for the length (top left panel), see fraction_1 in Table S4 above.



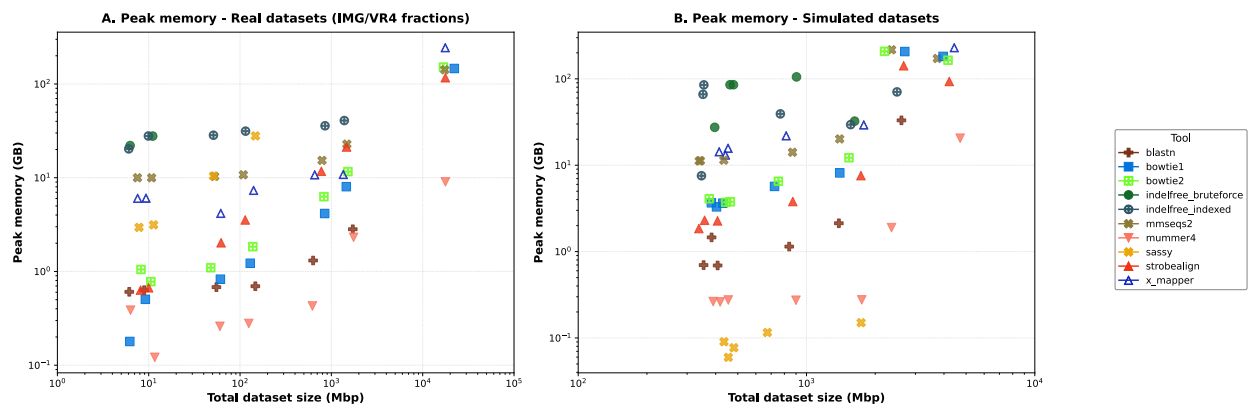
1.2.6 Supplementary Figure S6: Distribution of matched spacers from fraction_1

Distributions derived from the aggregate of all unique valid tool-reported alignments at hamming distance 3, from the results when scanning fraction_1 with the iPhoP spacer set. **A.** Length (bp) distribution of matched spacers. **B.** Total number of spacer occurrences (both axes are log-scaled), i.e. how many distinct contig regions each spacer aligned to. **C.** Number of reported unique alignments at each exact hamming distance value.



1.2.7 Supplementary Figure S7: Peak memory scaling (real and simulated datasets)

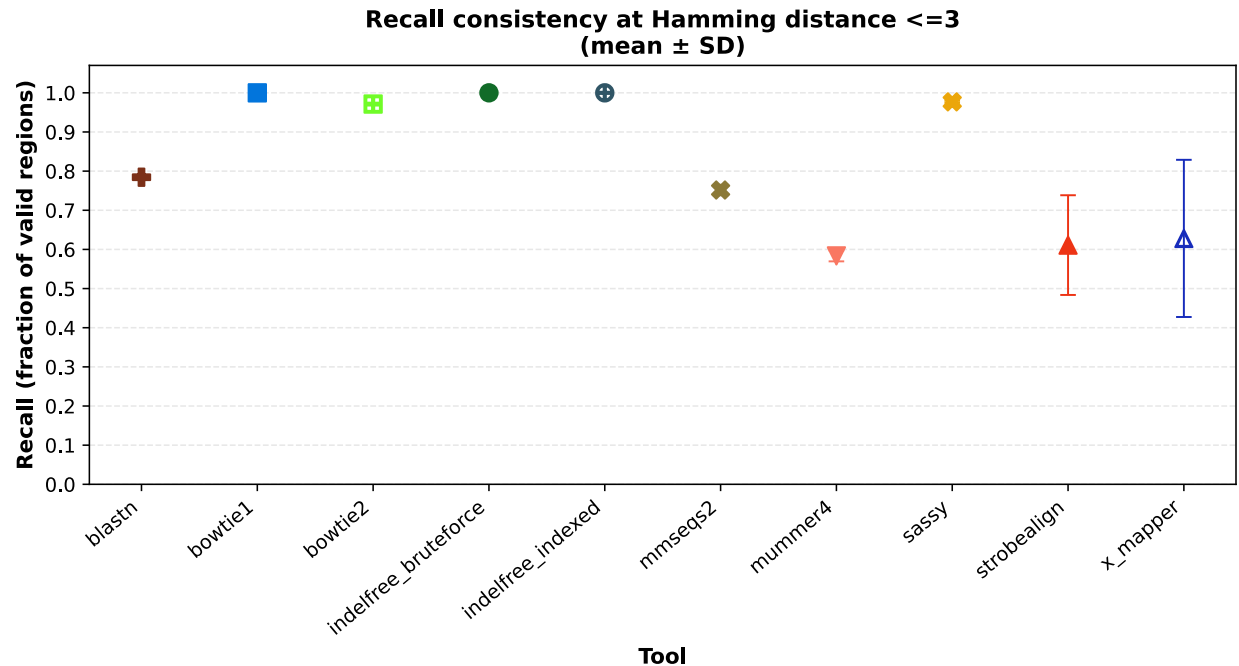
Peak memory usage (RSS, as reported by SLURM) scaling with dataset size. **A.** Real IMG/VR4 subsampled datasets (fractions 0.001-1.0). **B.** Simulated datasets with varying spacer and contig counts. Axes are log-scaled; tool identity is encoded by marker shape and color (see Table 1 in the main text for tool details).



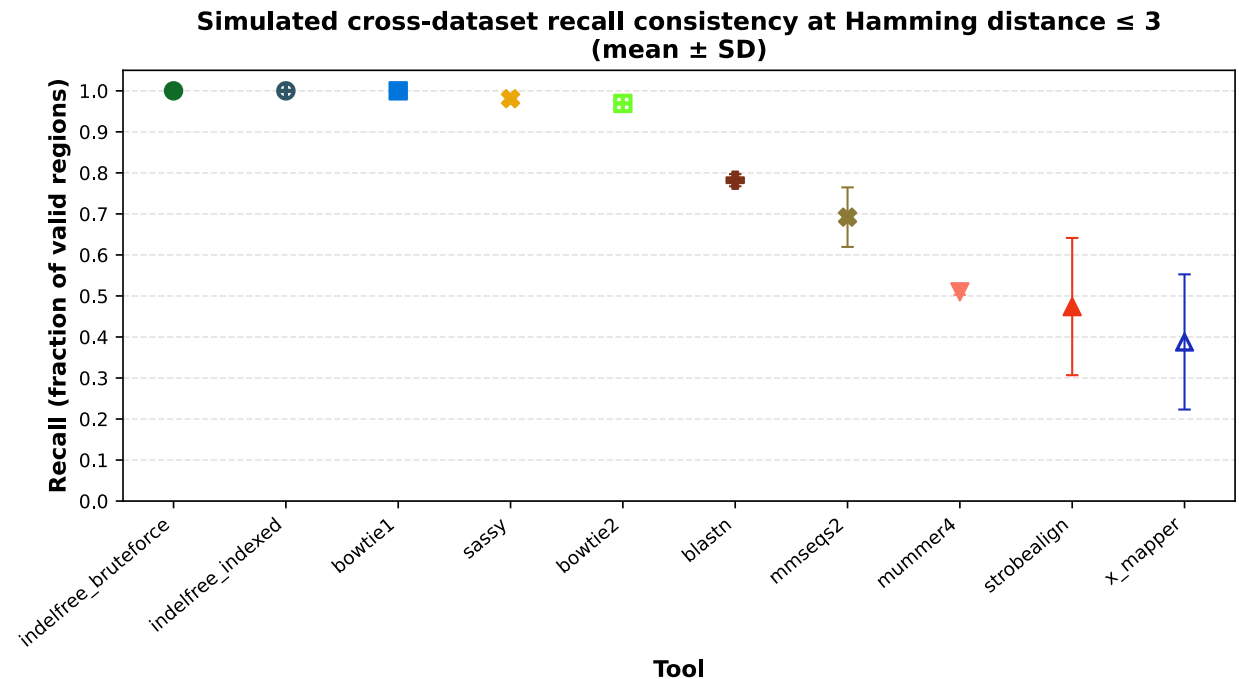
1.2.8 Supplementary Figure S8: Tool recall consistency across dataset sizes

Per-tool recall consistency at hamming distance 3. Upper panel: per-tool mean recall with error bars indicating standard deviation.

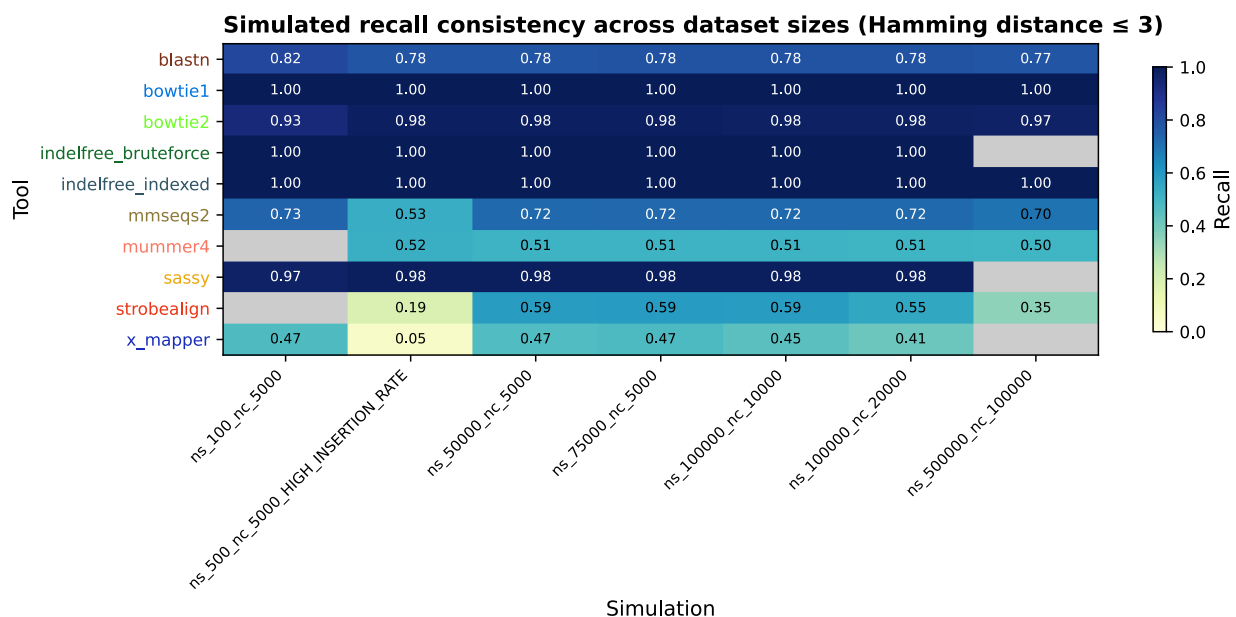
A. Real IMG/VR4 subsampled datasets.



B. Fully synthetic datasets.



C. Per-simulation (exact) recall values. Heatmap of per-tool recall values across simulated datasets at hamming distance 3. Rows correspond to tools and columns correspond to analyses runs.



1.3 Supplementary Notes

1.3.1 Supplementary Note 1: Coordinate Tolerance Matching Example

To illustrate the necessity for coordinate tolerance matching, consider the following case from one of the synthetic dataset (ns_100000_nc_10000) where three different tools detected the same ground truth region, but reported it with slightly different coordinates:

Source	Spacer ID	Contig ID	Query Start	Query End	Target Start	Target End	Strand	Mismatches/Cost (tool-reported)	Notes
Ground Truth	1c47e31b_1a47e31b0034	contig_1828		-	5001	5036	reverse	1	
BLASTn	1c47e31b_1a47e31b0034	contig_1828		34	5036	5003	reverse	0	
MMseqs2	1c47e31b_1a47e31b0034	contig_1828		35	5036	5002	reverse	1	
Sassy	1c47e31b_1a47e31b0034	contig_1828		-	5000	5036	reverse	1	CIGAR: 34=1I1=

In this example, BLASTn reports the ungapped alignment at contig positions 5003-5036 (0 reported mismatches), MMseqs2 reports 5002-5036 (1 reported mismatch over the terminal base), and Sassy reports 5000-5036 (1 reported insertion, suggesting the position 5000 in the contig matches that of the last position in the spacer). The ground truth planned occurrence was at 5001-5036. Despite these coordinate variations, all three tools detected the same biological match. **Critically**, all three tools report alignments that are acceptable matches to the ground truth occurrence, under a maximum hamming distance of 1.

This example demonstrates why a 5bp coordinate tolerance is necessary when aggregating results across tools to avoid double-counting essentially identical matches while accounting for valid algorithmic differences in gap versus substitution placement at alignment boundaries.

1.3.2 Supplementary Note 2: Non-planned Match Rate Analysis

1.3.2.1 Hamming vs edit distance: per-simulation results

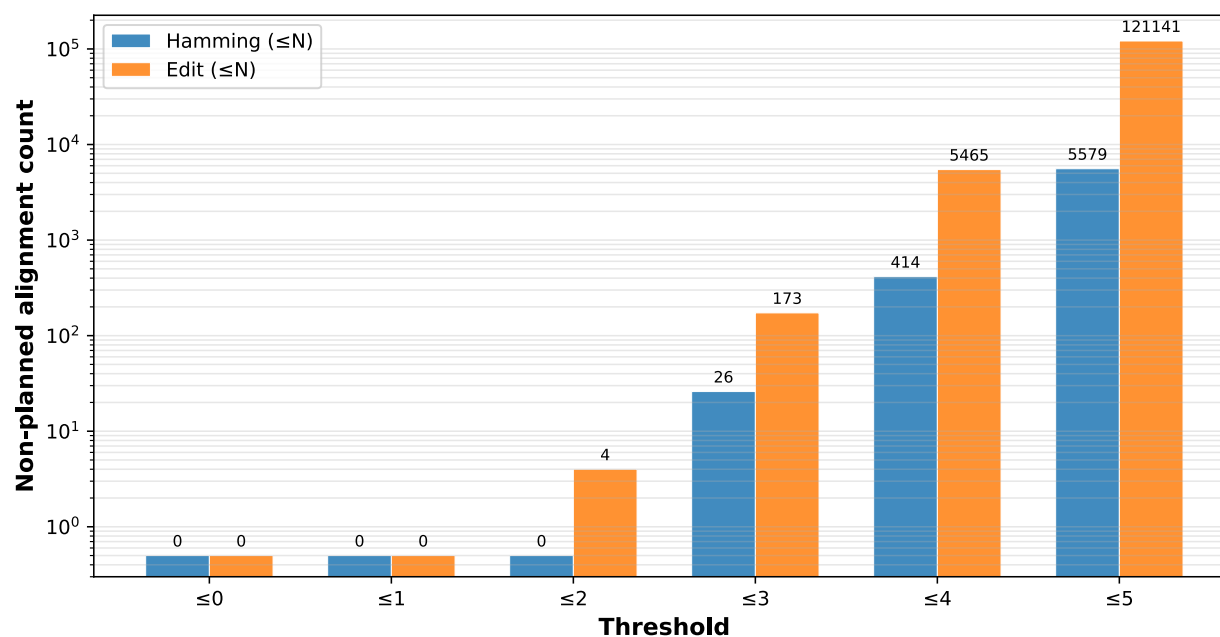
For each of the four fully synthetic simulations (ns_50000_nc_5000, ns_75000_nc_5000, ns_100000_nc_10000, ns_100000_nc_20000), we compared non-planned match counts under hamming distance (substitutions only) and edit distance (allowing indels) at thresholds 0 through 5.

1.3.2.1.1 Supplementary Figure S9: Per-simulation non-planned match counts (Hamming vs Edit)

Each panel below shows one simulation. Grouped bars compare hamming (blue) and edit (orange) distance non-planned match counts at each threshold (0–5, log scale). Annotations above each bar indicate exact counts.

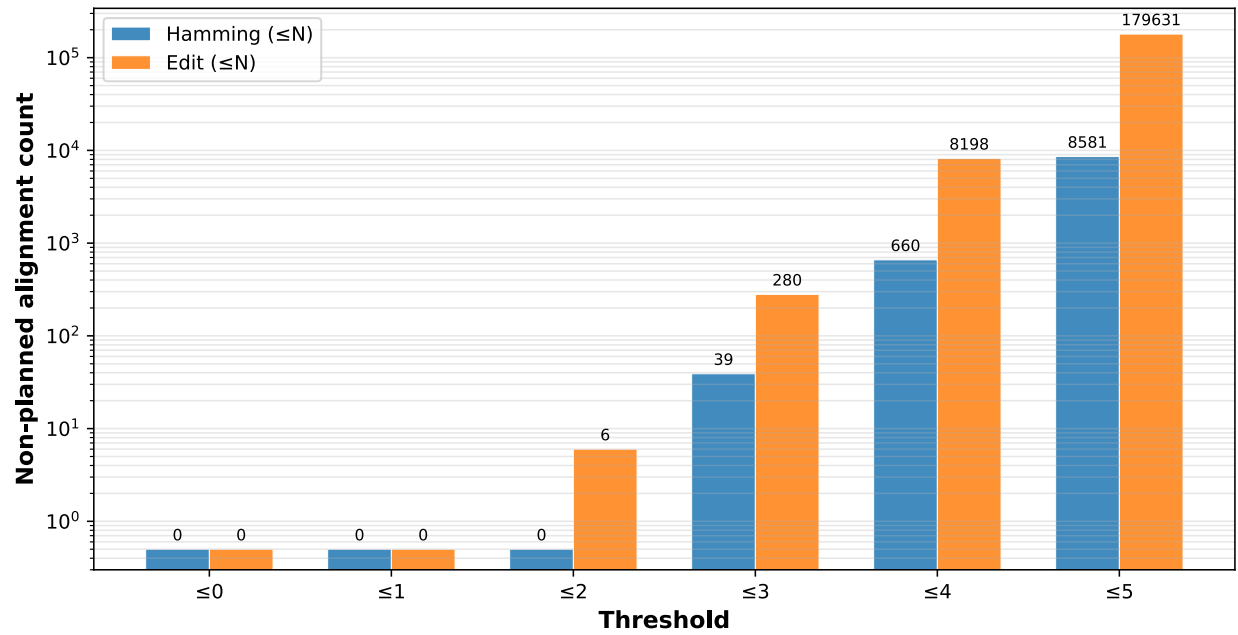
A. ns_50000_nc_5000: 50,000 spacers (1.6 Mbp), 5,000 contigs (403 Mbp)

ns_50000_nc_5000: 50,000 spacers × 5,000 contigs



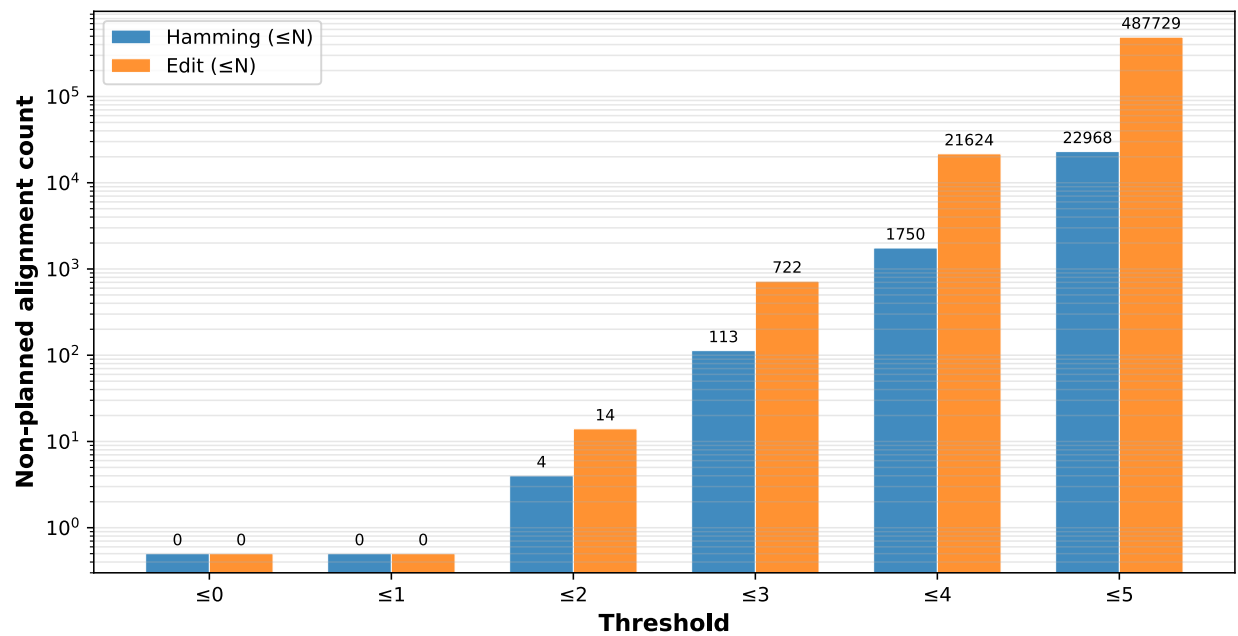
B. ns_75000_nc_5000: 75,000 spacers (2.4 Mbp), 5,000 contigs (401 Mbp)

ns_75000_nc_5000: 75,000 spacers × 5,000 contigs



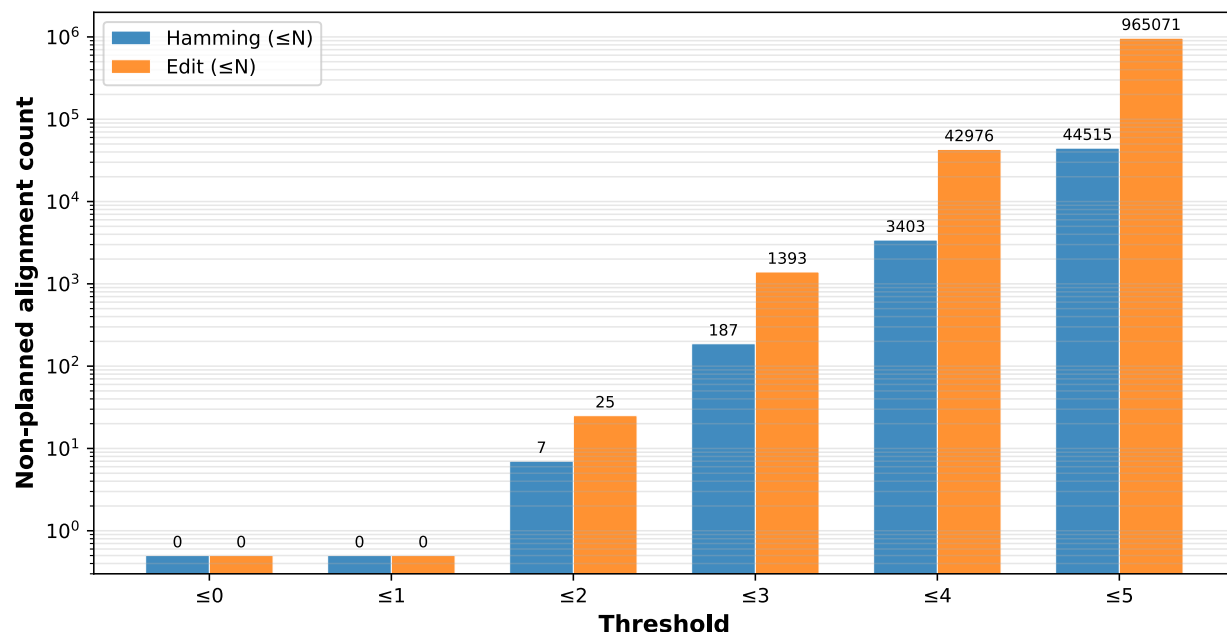
C. ns_100000_nc_10000: 100,000 spacers (3.2 Mbp), 10,000 contigs (803 Mbp)

ns_100000_nc_10000: 100,000 spacers × 10,000 contigs



D. ns_100000_nc_20000: 100,000 spacers (3.2 Mbp), 20,000 contigs (1.6 Gbp)

ns_100000_nc_20000: 100,000 spacers × 20,000 contigs



1.3.2.2 Search-space-normalized non-planned match rates

1.3.2.2.1 Supplementary Table S7: Search-space-normalized metrics at hamming/edit distance 3

The table below summarizes the normalized non-planned match rates at distance threshold 3 for each simulation, extracted from the full dataset in `search_space_normalized_metrics.csv`. The threshold 3 subset is also available as `search_space_normalized_metrics_threshold3.csv`.

Table 13: Non-planned match rates at distance threshold 3 for each fully synthetic simulation. The per Gbp² metric is consistent across simulations for both hamming (approximately 40,000) and edit (approximately 270,000–290,000) distance, confirming that this metric appropriately normalizes for search space size. Edit distance 3 produces approximately 6–8 fold more non-planned matches than hamming distance 3.

Simulation	Metric	n_spacers	contig_bp (Gbp)	Non-planned	Per spacer per Gbp	Per Gbp ²
ns_50000_nc_50000	Hamming	50,000	0.40	26	0.0013	40,371
ns_50000_nc_50000	Edit	50,000	0.40	173	0.0086	268,621
ns_75000_nc_50000	Hamming	75,000	0.40	39	0.0013	40,489
ns_75000_nc_50000	Edit	75,000	0.40	280	0.0093	290,692
ns_100000_nc_100000	Hamming	100,000	0.80	113	0.0014	43,968
ns_100000_nc_100000	Edit	100,000	0.80	722	0.0090	280,929
ns_100000_nc_20000	Hamming	100,000	1.60	187	0.0012	36,427
ns_100000_nc_20000	Edit	100,000	1.60	1,393	0.0087	271,352

1.3.2.2.2 Supplementary Table S8: Semi-synthetic non-planned match rates by hamming threshold

Table 14: Non-planned matches in the semi-synthetic dataset (3,826,979 real spacers $\times$ 421,431 simulated contigs, 0 planned spacer “injections”) at cumulative hamming distance thresholds. The per Gbp² rate at hamming 3 (101,270) is higher than observed in the fully synthetic datasets (approximately 40,000), which is expected given differences in sequence composition between real and simulated spacers (e.g., non-uniform k-mer distribution, biological sequence features). Values at thresholds 4 and 5 are **lower bounds** as only heuristic tools were used. {#tbl-supp-semisynthetic-rates} The comparison of normalized non-planned match rates between the semi-synthetic dataset and the mean across fully synthetic simulations at hamming distance 3 is presented in the main text (Table 3 in the main text).

Hamming threshold	Non-planned matches	Per spacer per Gbp	Per Gbp ²
0	1	6.1×10^{-8}	1.80
1	49	3.0×10^{-6}	88.2
2	2,354	1.4×10^{-4}	4,236
3	56,273	3.4×10^{-3}	101,270
4	72,132	4.4×10^{-3}	129,810
5	113,314	6.9×10^{-3}	203,922

1.3.2.3 Caveats and interpretation

These non-planned match rate estimates should be interpreted qualitatively rather than as exact predictive values. Real spacer sequences have compositional features (conserved motifs, non-random k-mer structure, locus-specific GC variation) not fully captured by simulated sequences, as illustrated by the approximately 2.5-fold difference between semi-synthetic and fully synthetic rates. Simulated contigs, while matched to IMG/VR4 in GC content and length distributions, lack additional structural features of real viral genomes (coding density, gene arrangement, horizontal transfer footprints) that may influence match frequencies. For the semi-synthetic dataset, only heuristic tools were computationally feasible, so the reported counts represent lower bounds on the true non-planned match frequency. These estimates are most informative for relative comparisons (hamming vs edit distance, threshold effects, and scaling with search space size); for absolute prediction of non-planned match counts in a given analysis, they should be treated as order-of-magnitude guides.

Complete analysis available in: `distance_metric_analysis_v3.ipynb` notebook in project repository, including per-distance breakdowns, tool-specific validation analyses, and detailed comparison tables.

1.4 References and Data Availability

All analysis notebooks, source code, and generated data are available in the project repository at: github.com/UriNeri/spacer_matching_bench.

Key analysis notebooks referenced in supplementary materials: - `spacer_inspection.ipynb`: Complexity filtering, spacer feature analysis, simulated vs real comparison - `distance_metric_analysis.ipynb`: Non-planned match rate analysis, coordinate tolerance examples, hamming vs edit distance comparison - `Prepare_all_jobs.ipynb`: Simulation parameter definitions, dataset generation commands - `Performance_simulated_v2.ipynb`: Per-tool performance metrics on simulated datasets - `Performance_imgvr4_v3.ipynb`: Per-tool performance metrics on real IMG/VR4 datasets

Dataset availability: - Filtered IMG/VR4 HQ contig subset (421,431 contigs, ~18.9 Gbp): [Zenodo DOI] - Filtered iPhoP spacer dataset (3,826,979 spacers, 132 Mbp): [Zenodo DOI] - All synthetic datasets and ground truth files: [1]

References

- [1] U. Neri et al. “Supplementary data for CRISPR spacer-protospacer matching benchmarks [Data set]”. In: (). DOI: [10.5281/zenodo.15171878](https://doi.org/10.5281/zenodo.15171878).